

Flare Summer Signal

Grand Prize Strategy Report

Evidence-based blueprint for maximising win probability on Flare

Hackathon	Flare Summer Signal (DoraHacks, virtual)
Organiser	Flare Networks
Programme	June 29 – August 14, 2026 (final submission)
Prize pool	USD 12,000 across two bounties (plus Grand Prize)
Tracks	Bounty 1: Interoperable Asset Products · Bounty 2: Confidential Compute Apps
Registered	~415 hackers at time of research
Core stack	FAssets (FXRP), FDC, FTSO, Flare Confidential Compute (FCC), PMW

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Every significant claim in this report is tagged with one of the three evidence levels above. Where information is unavailable, the uncertainty is stated explicitly. No strategy can guarantee victory; the objective is to maximise the probability of producing a project that stands out through exceptional research, technical execution, product thinking, and alignment with Flare’s goals.

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Executive Summary

This report presents the findings of a multidisciplinary research effort whose objective is to identify the project with the highest probability of winning the Grand Prize of the **Flare Summer Signal** hackathon hosted on DoraHacks. The analysis covers the hackathon's official rules and judging criteria, the strategic priorities of the Flare Networks organisation, the historical record of prior Flare hackathons, the implicit psychology of the judging panel, the broader competitive landscape, and a structured discovery process that produced and scored more than fifty candidate concepts before converging on a single recommended project.

The hackathon is a six-week online programme running from 29 June to 14 August 2026, with two parallel bounties of USD 6,000 each — one for Interoperable Asset Products and one for Confidential Compute Apps — plus a Grand Prize that recognises the best overall submission across both tracks. The judging rubric is unusually explicit and concentrates on five dimensions: product usefulness, Flare integration quality, technical execution, evidence of new work, and clarity with future potential. The single most heavily weighted dimension, in our assessment, is **Flare integration quality**: judges are explicitly asked to distinguish between meaningful and superficial use of Flare's protocols.

Three structural facts about the Flare ecosystem shape the strategic recommendation. First, Flare Confidential Compute (FCC) — Flare's TEE-based verifiable compute layer and the first delivery of the Flare 2.0 vision — is brand new, having only just received governance approval to launch on the Songbird canary network in July 2026. It has almost no live applications. Second, Protocol Managed Wallets (PMW), the cross-chain execution primitive that extends Flare's consensus to external chains such as XRPL and Bitcoin, is even newer and similarly devoid of real applications. Third, Flare's commercial momentum is now institutional — VivoPower's USD 100 million XRP treasury strategy, BitGo's regulated custody for FLR, and Kraken's integration of USDT0 on Flare all point to a network that is actively courting institutional capital but lacks the institutional-grade product surface to absorb it.

The intersection of these three facts produces a narrow but exceptionally strong strategic opening: **build the first credible institutional-grade product that uses FCC and PMW together to give XRP-native treasury holders a verifiable, private, AI-managed cross-chain asset management layer**. This is the concept this report develops in full under the working name **Aegis** — A Verifiable, Confidential, AI-Managed Cross-Chain Treasury and Autonomous Risk Layer for XRP-Native Institutions on Flare.

Aegis uses every enshrined Flare primitive in a load-bearing way: FAssets (FXRP) as the core asset, FTSO for price discovery and risk signals, FDC for cross-chain state and payment attestation, FCC for confidential position computation and verifiable AI inference, and PMW for cross-chain execution across Flare, XRPL, Base, and Hyperliquid. It targets a real and growing customer segment — corporate XRP treasuries and large FLR holders — that has been publicly validated by the VivoPower partnership. It has a defensible business model (management and performance fees, plus an institutional SaaS tier). It demos exceptionally well: the live demo can show a private vault, an AI agent rebalancing positions inside a TEE, cross-chain execution via PMW, and a zero-knowledge-style solvency proof verifiable by an external auditor — all in under five minutes.

Headline recommendation

Build **Aegis**. Win Bounty 2 (Confidential Compute Apps) outright and place strongly enough in Bounty 1 (Interoperable Asset Products) to capture the Grand Prize. The project dominates the judging criterion that matters most — Flare integration quality — by being the only submission that uses FCC, PMW, FDC, FTSO, and FAssets together in non-trivial ways. It also delivers against every other criterion: real product usefulness (institutional treasury management), strong technical execution (a working TEE extension and smart contracts on Coston2), credible evidence of new work, and a clear post-hackathon commercial path.

At-a-glance: why Aegis wins

Judging criterion	How Aegis performs	Edge vs. field
Product usefulness	Solves a real institutional pain point validated by Vivo and Bitfarms partnerships	Most other teams are demos; few address institutional capital.
Flare integration quality	Uses FCC + PMW + FDC + FTSO + FAssets, each loaded with data	Likely the only entry combining all five primitives.
Technical execution	Working FCC extension, vault contracts, PMW-mediated cross-chain deposits	Demonstrably deeper than typical FCC "Hello World" ports.
Evidence of new work	All core components newly built during the six-week window	Avoids the "ported existing project" dilution risk.
Clarity & future potential	Clear ICP, business model, and post-hackathon roadmap	Stronger overall narrative than research-only entries.

1. Hackathon Intelligence Report

This section consolidates everything that is publicly knowable about the Flare Summer Signal hackathon from official sources, the DoraHacks listing, the Flare Developer Hub, Flare's news feed, and related ecosystem communications. Each claim is tagged with an evidence level. Where sources conflict, the discrepancy is flagged rather than smoothed over.

1.1 Programme structure and timeline

VERIFIED FACT Flare Summer Signal is an open, virtual hackathon hosted on DoraHacks. Registration and development open on 29 June 2026; the final submission deadline is 14 August 2026; judging runs 15–21 August; winners are announced on 24 August 2026. The programme therefore gives builders approximately six and a half weeks of build time — substantially longer than the typical 48–72 hour in-person hackathon and long enough to produce a credible production-grade MVP.

VERIFIED FACT The DoraHacks detail page lists two bounties. **Bounty 1 — Interoperable Asset Products** has a prize pool of USD 6,000 (USD 4,000 for first place, USD 2,000 for second). **Bounty 2 — Confidential Compute Apps** has the same structure and the same payouts. The headline prize pool on the listing is therefore USD 12,000. A separate Flare post on X describes the event as offering six weeks, USD 12K in prizes, and two tracks; a third-party summary references a total prize pool of USD 21,000 including an additional Grand Prize component. We treat the USD 12,000 figure as Verified Fact (it appears on the official DoraHacks page) and the existence of a separate Grand Prize as Evidence-Based Inference (multiple independent references mention it, but it is not described on the official page).

INFERENCE The most important implication of the structure is strategic: a single project that is competitive in *both* bounties simultaneously has a strictly larger expected payout than one targeting a single bounty, because it can win its primary bounty outright while also being eligible for the Grand Prize. This is a non-trivial consideration given that the two tracks map directly onto Flare's two flagship technology bets — FAssets (interoperability) and FCC (confidential compute). A project that fuses both is also the most legible demonstration of the Flare 2.0 thesis, which is precisely the story the organisers want to tell the market.

1.2 Submission requirements

VERIFIED FACT Submissions must include: project name; selected bounty or bounties; short product description; target user; demo link, video, or working app link; GitHub repo or technical materials; an explanation of how the project uses Flare; an explanation of what was newly built, ported, integrated, or improved during the programme; smart contract addresses or deployment details where applicable; and a short roadmap or next steps. Teams are also encouraged (but not required) to disclose whether they deployed on Coston2, Songbird, or Flare Mainnet, and to share any early traction signals — pilot users, partner conversations, community interest, or real user feedback.

INFERENCE The submission requirements are unusually commercial in tone. The explicit ask for traction signals, pilot users, and partner conversations signals that the judges will reward projects with a credible post-hackathon trajectory over technically impressive but commercially inert demos. Existing projects are explicitly permitted, provided teams clearly separate prior work from new work. This is a meaningful departure from the typical hackathon framing and should shape how the submission is written: a clean before/after narrative of what the Flare integration unlocked is more valuable than a feature list.

1.3 Judging criteria — verbatim and decoded

VERIFIED FACT The official judging criteria, as published on the DoraHacks detail page, are: (1) **Product usefulness** — does the product solve a real user, developer, ecosystem, or infrastructure problem? (2) **Flare integration quality** — is Flare used in a meaningful way, or is the integration superficial? (3) **Technical execution** — does the demo work, and is the architecture credible and understandable? (4) **Evidence of new work** — did the team clearly show what was newly built, ported, integrated, or improved? (5) **Clarity and future potential** — can the team explain the product, user, integration, and next steps clearly, and does the project have a credible path beyond the hackathon?

INFERENCE Decoding the rubric, the second criterion — Flare integration quality — is the single most powerful discriminator. It is the only criterion framed as a binary test ("meaningful or superficial"), and it is the criterion most directly aligned with the organiser's strategic interest in proving that Flare's unique primitives are useful. A project that uses FCC in a load-bearing way (i.e., the product could not exist without it) will score dramatically higher on this criterion than a project that uses FCC cosmetically (e.g., a single TEE call grafted onto an otherwise ordinary dApp). This is the single most important design constraint in the entire hackathon, and it drives the recommendation in this report.

1.4 Sponsors, organisers, and partners

VERIFIED FACT The hackathon is organised directly by Flare Networks. The DoraHacks listing does not name a separate headline sponsor; Flare is both organiser and funder. The tag set on the listing — Blockchain, AI, Platform technology, Flare — confirms the deliberate positioning of Flare as a platform technology with AI as a first-class use case. The Flare Hackathon Telegram group and the Flare Developer Hub (dev.flare.network) are the official support channels.

VERIFIED FACT Flare's recent commercial partnerships, as documented in the network's third-anniversary summary and subsequent news flow, include Google Cloud (Confidential Compute co-development and the prior Verifiable AI hackathon), BitGo (regulated custody for FLR and SGB in the US and Europe, June 2025), VivoPower (a USD 100 million XRP treasury strategy, June 2025), EasyA (developer onboarding series for XRPL builders, June 2025), Kraken (integration of USDT0 on Flare, April 2025), and Hyperliquid (the FXRP/USDH spot market). These partnerships are not formal sponsors of the Summer Signal hackathon, but they define the commercial context the organisers care about.

INFERENCE The partnership roster reads as a deliberate institutional onboarding strategy: custody (BitGo), treasury demand (VivoPower), stablecoin liquidity (Kraken, USDT0), developer pipeline (EasyA), and an AI infrastructure partner (Google Cloud). Projects that visibly serve one or more of these institutional tracks are more likely to be perceived by the organisers as strategically useful, which is a material advantage in a judge's mind even if it does not appear explicitly in the rubric.

1.5 Useful links and developer resources

VERIFIED FACT The official Flare Developer Hub at dev.flare.network is the primary technical resource. It contains dedicated documentation for FAssets, the Flare Data Connector (FDC), the Flare Time Series Oracle (FTSO), and Flare Confidential Compute (FCC), including a 'Build your first extension' quickstart that walks through deploying a Go HTTP server inside a TEE on the Coston2 testnet. The Flare Foundation GitHub organisation (github.com/flare-foundation) hosts the FDC suite deployment repository, the flare-ai-skills repository (which includes a flare-fdc-skill with structured guidance on attestation types, request flow, Merkle proofs, and supported networks), and the FCC scaffold example referenced by the quickstart.

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The FDC supports attestations from Bitcoin, Dogecoin, and the XRP Ledger (non-smart-contract chains), and from Ethereum, Songbird, and Flare (smart-contract chains). Supported attestation types include EVMTransaction, Web2Json, Payment, AddressValidity, XRPPayment, and XRPPaymentNonexistence — a deliberately broad surface that enables proof-of-reserves, weather insurance, cross-chain payment verification, and arbitrary Web2 data attestation.

2. Reverse-Engineering the Organisers

The published rules and rubric tell us what the organisers will measure. To understand what they actually want, we have to read their behaviour: what they have chosen to build, what they have chosen to fund, what they have chosen to promote, and what they have chosen to launch on the eve of this hackathon. This section separates what we can verify from what we infer.

2.1 The future Flare is trying to accelerate

- VERIFIED FACT** Flare's self-description, both on the hackathon page and across its news feed, is consistent: an EVM-compatible Layer 1 'built to unlock DeFi for assets that do not have native smart contracts, starting with XRP through FAssets,' and 'the blockchain for data.' The Flare 2.0 vision, published on 19 March 2025, extends this to a broader claim: Flare's consensus should be extended beyond its own chain by combining Trusted Execution Environments with Flare's existing data protocols.
- INFERENCE** The future Flare is trying to accelerate is one in which (a) XRP and other non-smart-contract assets are first-class citizens of EVM DeFi, (b) verifiable off-chain compute — particularly AI — is a native capability rather than a bolt-on, and (c) Flare becomes the coordination layer for cross-chain execution across both EVM and non-EVM chains without relying on bridges. The hackathon's two bounties map directly onto bets (a) and (b); bet (c) — PMW — is the connective tissue between them.

2.2 The ecosystem gaps that appear most urgent

- VERIFIED FACT** FCC was first proposed as research in 2024, the architecture was published with the Flare 2.0 vision in March 2025, and the protocol went to a Songbird governance vote that ran 6–13 July 2026. As of late July 2026, FCC is 'still in the final development phases and available on the Coston2 test network,' in the words of a Flare developer. PMW launched with the same governance vote with initial support for XRPL.
- INFERENCE** The most urgent gap, by a wide margin, is the absence of any production application that exercises FCC and PMW together. These are the two newest and most strategically important primitives in the Flare stack, and they have no live demonstrations of meaningful use. A hackathon submission that credibly uses both is, in effect, a reference implementation that Flare can point to in subsequent marketing and investor communications. This is an extraordinarily high-leverage position to occupy.
- INFERENCE** The second urgent gap is institutional-grade XRPFi product surface. The current XRPFi stack is retail-oriented: Sceptre for liquid staking, Blazeswap for AMM swaps and LP, Kinetic for lending, Firelight for XRP staking, and Clearstar for XRP yield vaults. There is no product that an institution like VivoPower could plausibly use to manage a USD 100 million XRP treasury with confidentiality, cross-chain execution, and verifiable risk controls. The gap between Flare's institutional partnerships and its institutional product surface is widening.
- INFERENCE** The third gap is verifiable AI done seriously. The prior Flare x Google Cloud Verifiable AI Hackathon produced several impressive demos — 2DeFi, Flare Fact Checker, Command Flare, NFT Deep Appraisal, Quince Finance, ScribeChain — but all were hackathon-grade demonstrations. None has become a production product. The organisers have a strong incentive to see at least one verifiable AI concept mature into something with a real user and a real revenue line.

2.3 The kinds of startups Flare would likely fund after the hackathon

VERIFIED FACT Flare operates an active grants programme (referenced as 'Grants' in the Developer Hub navigation). The Flare Foundation GitHub organisation maintains open-source infrastructure (FDC suite deployment, flare-ai-skills). The network's commercial partnerships with BitGo, VivoPower, and Hyperliquid are not grants, but they indicate the kinds of companies Flare's business development team is willing to engage with: regulated custodians, treasury-heavy corporates, and venues that create FXRP pair liquidity.

INFERENCE Synthesising these signals, the startups Flare is most likely to fund or accelerate after the hackathon fall into three clusters. First, infrastructure that makes FCC and PMW usable for other developers — SDKs, attestation marketplaces, TEE orchestration layers. Second, institutional DeFi products that absorb the demand created by the VivoPower-style treasury partnerships — private vaults, compliance layers, regulated yield products. Third, verifiable AI applications that turn Flare's data protocols into a defensible moat for AI workloads — verifiable inference, confidential agents, federated learning coordinators.

2.4 Which developer behaviours the organisers are encouraging

INFERENCE Three behaviours are clearly being encouraged by the structure of the programme. First, **shipping on Flare rather than on a test slide**: the explicit ask for smart contract addresses and deployment details, and the optional disclosure of whether the project is on Coston2, Songbird, or Mainnet, signal that the organisers want real deployments, not whitepapers. Second, **building seriously rather than porting lightly**: the rules' careful separation of prior work from new work, and the explicit criterion of 'evidence of new work,' discourage the common hackathon tactic of grafting a thin Flare integration onto an existing product. Third, **thinking commercially rather than only technically**: the invitation to share traction signals and partner conversations is a strong signal that the organisers want builders who are thinking about distribution, not just code.

2.5 Which technologies are strategic priorities

VERIFIED FACT Flare's three enshrined protocols — FTSO, FDC, and FCC — are described on the Developer Hub as 'enshrined (FSP) into the network, inheriting the economic security of the entire chain.' FAssets and Flare Smart Accounts are the asset layer on top. PMW is the cross-chain execution layer enabled by FCC. Verifiable AI is the application layer that the Flare 2.0 deck identifies as the third strategic pillar: 'Agents run in confidential compute with guardrails; execution routes via PMW across Flare, XRP, ETH, or traditional rails.'

INFERENCE The strategic priority stack is therefore (from lowest to highest layer): FAssets and Smart Accounts (asset onboarding), FTSO (price data), FDC (state and event attestation), FCC (verifiable confidential compute and PMW), and verifiable AI (the application layer). A project that touches all five layers is maximally aligned with Flare's stated technology strategy — and is also maximally difficult to build, which is precisely why it is scarce and valuable.

2.6 Which types of projects consistently receive attention

VERIFIED FACT The Verifiable AI Hackathon winners spotlight published on flare.network on 26 March 2025 highlighted six projects across two editions. The in-person winners were 2DeFi (AI x DeFi 1st, RAG 2nd), Flare Fact Checker (Social AI Agents 1st), Command Flare (RAG 1st, Consensus Learning 3rd), and NFT Deep Appraisal (Consensus Learning 1st). The virtual winners included Quince Finance (AI x DeFi 1st) and ScribeChain (Social AI Agents 1st).

INFERENCE

The pattern is clear. Projects that received attention combined three characteristics: (a) they used Flare's data protocols (FTSO, FDC) or TEEs non-trivially, (b) they had a polished UX that made the demo immediately intelligible (Google login, voice commands, conversational interfaces, Farcaster integration), and (c) they solved a problem that was legible to a non-technical observer (onboarding to DeFi, fact-checking, NFT appraisal). Technical depth alone was not sufficient; the projects that won were the ones whose value proposition could be understood in one sentence.

2.7 Which ideas are overrepresented

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Three idea categories are likely to be overrepresented in the Summer Signal submission pool. First, AI-powered onboarding assistants in the style of 2DeFi and Quince Finance — the prior winners have established a template (Google login, embedded wallet, conversational DeFi) that will be widely copied. Second, generic RAG-over-Flare-docs chatbots — the RAG Knowledge track of the prior hackathon attracted many such submissions and the pattern persists. Third, simple FXRP wrappers — basic lending, staking, or AMM vaults that use FAssets but do not exercise FDC, FCC, or PMW. These will be common in Bounty 1 and will score poorly on Flare integration quality because the FAssets integration is the only one present.

2.8 Which demos generate emotional excitement

INFERENCE

Based on the prior winners' descriptions and the language used to promote them, the demos that generate the most emotional excitement are those that produce a visible, surprising result live. 2DeFi's ability to scan a Robinhood portfolio from a single photograph and produce a Flare DeFi strategy is the canonical example: the wow moment is the photograph-to-strategy transformation. Quince Finance's voice-controlled DeFi ('Swap 50 FLR for SGB, hands free') is another. The common pattern is a demo in which an opaque input (a photo, a voice command, a natural language instruction) produces a verifiable, on-chain action — and the verification step is shown explicitly.

INFERENCE

The implication for the Summer Signal is that the demo should include at least one moment in which a sensitive or complex input (e.g., an institution's complete treasury position) is processed confidentially inside a TEE and produces an on-chain action (e.g., a cross-chain rebalance via PMW) that can be independently verified. The confidentiality-to-verifiability transformation is the emotional core of the Flare 2.0 thesis, and a demo that dramatises it will land harder than any feature list.

2.9 Which engineering achievements consistently impress judges

INFERENCE

Across both the prior Verifiable AI Hackathon and the broader Flare ecosystem, the engineering achievements that impress judges are: (a) a working TEE deployment with real attestation (not a mocked TEE), (b) the use of multiple Flare primitives in combination rather than in isolation, (c) a credible deployment on a real Flare network (Coston2 minimum, Songbird preferred), and (d) an AI component that is actually doing useful work inside the TEE rather than calling an external LLM API and presenting the response as 'AI.' The difference between a real TEE deployment and a mocked one is usually visible in the attestation proof flow, and judges with technical depth will look for it.

2.10 What distinguishes unforgettable projects



Unforgettable projects combine four qualities that are rarely present together. First, a **thesis-level insight**: a one-sentence articulation of why this product must exist on this chain at this moment. Second, a **structural use of the chain's unique primitives**: the product could not be built on any other chain because it depends on something only this chain provides. Third, a **demo that dramatises the thesis**: the wow moment is the thesis made visible. Fourth, a **credible post-hackathon trajectory**: the team has a clear next step and ideally an early signal of demand. Projects that hit all four are rare and memorable; projects that hit three of four are common and forgettable.

3. Ecosystem Landscape Analysis

This section maps the Flare ecosystem as it stands in late July 2026: the existing products, the strategic infrastructure, the known gaps, and the commercial context. The objective is to identify where the leverage for a new project is highest.

3.1 Existing products and the current stack

VERIFIED FACT The Flare DeFi stack is concentrated around a small number of protocols. **Sceptre** is the dominant liquid staking protocol, issuing sFLR receipt tokens that accrue staking and FlareDrop rewards while remaining composable in DeFi; it is built by Rome Blockchain Labs. **Blazeswap** is the primary AMM and liquidity venue for sFLR pairs. **Kinetic** is a lending and borrowing protocol from the same team as Sceptre, designed to work with sFLR. **Firelight** issues stXRP, a liquid staking token for XRP, by integrating with Flare's FAssets system to mint FXRP and then stake it. **Clearstar** operates the Flare XRP Yield Vault referenced in the Smart Accounts v1.3 release. **Enosys** is a Flare-native DEX. **Digital Dynamix** operates an NFT marketplace on Flare.

VERIFIED FACT Flare's native reward mechanisms are threefold: staking FLR to infrastructure providers (which supports consensus and data provision), delegating FLR to FTSO data providers (which supports decentralized price feeds and earns delegation rewards), and receiving monthly FlareDrops. Sceptre's value proposition is that it consolidates these three reward streams into a single composable receipt token.

3.2 Strategic infrastructure — the three enshrined protocols

3.2.1 Flare Time Series Oracle (FTSO)

VERIFIED FACT The FTSO is Flare's decentralized price feed. Data providers submit prices in vote-rounds; a weighted median determines the canonical price; providers are rewarded for accuracy. The FTSO is enshrined, meaning its economic security is inherited from the Flare chain itself rather than from a separate token. As of the FTSO deep-dive workshop hosted by Encode Club in January 2025, the FTSO supports a broad set of price pairs and is the canonical source of on-chain price data for Flare DeFi.

3.2.2 Flare Data Connector (FDC)

VERIFIED FACT The FDC is an enshrined oracle for arbitrary external data. It uses a Request-Commit-Reveal protocol in which attestation providers verify data from external blockchains or Web2 APIs; verified data is stored in a Merkle tree with only the root published on-chain. Supported source chains include Bitcoin, Dogecoin, the XRP Ledger, Ethereum, Songbird, and Flare. Supported attestation types include EVMTransaction, Web2Json, Payment, AddressValidity, XRPPayment, and XRPPaymentNonexistence. The flare-ai-skills repository's flare-fdc-skill explicitly lists weather insurance, proof-of-reserves, and cross-chain payment verification as canonical use cases.

3.2.3 Flare Confidential Compute (FCC)

VERIFIED FACT FCC is the first delivery of the Flare 2.0 vision. It combines Trusted Execution Environments (hardware enclaves that isolate computation from the host machine) with Flare's data protocols. Two pillars launched with the Songbird governance vote in July 2026: **Protocol Managed Wallets (PMW)**, which give protocols on Flare the ability to create and operate wallets on external chains (initially XRPL) controlled by protocol rules rather than by a backend server, with keys generated and held inside TEEs that sign only when a weighted majority of data providers relays the instruction; and a **compute extension** that allows general-purpose verifiable off-chain computation. The Developer Hub's quickstart demonstrates a Go HTTP server running inside a TEE that receives instructions from an on-chain InstructionSender smart contract and returns results that callers poll from an extension proxy.

3.3 Asset layer — FAssets, FXRP, and Smart Accounts

VERIFIED FACT FAssets is the system that mints 1:1 representations of non-smart-contract assets on Flare. **FXRP** is the first live FAsset: a 1:1 representation of XRP on Flare, backed by over-collateralization, that makes XRP usable across EVM DeFi while remaining connected to XRPL as the native settlement layer. FAssets v1.3, and the separate FXRP OFT route, handle cross-network transfers between Flare and supported networks such as Base, HyperEVM, HyperCore, and Monad. The Flare Smart Accounts (FSA) v1.3 release enables XRP holders to mint FXRP and deposit into yield-generating vaults using a single XRPL wallet signature, with support for Ledger, Xaman, and Joey Wallet.

VERIFIED FACT Flare's commercial framing of FXRP is direct: more than 150 million FXRP is in circulation, with roughly 85% deployed across DeFi. The Hyperliquid FXRP/USDH spot market creates a venue for FXRP pair trading. The VivoPower partnership demonstrates corporate treasury demand for XRP exposure managed through Flare.

3.4 Tokenomics and value accrual

VERIFIED FACT Flare's tokenomics proposal, documented in 'From activity to value accrual: Flare's plan for FLR,' cuts annual inflation from 5% to 3%, introduces protocol-level MEV capture, and routes a broader set of network revenues through the Flare Income Reinvestment Entity (FIRE) toward supply reduction and ecosystem growth. The base gas fee was raised to 1200 gwei, targeting burns of up to approximately 300 million FLR annually. A separate FIP.01 burn retired approximately 198.9 million FLR immediately, with further monthly burns scheduled through January 2026.

INFERENCE The tokenomics design has a clear strategic intent: turn network activity into FLR value accrual. This means the organisers have a direct economic interest in projects that generate on-chain activity — particularly activity that flows through FAssets minting (which generates protocol fees), Smart Accounts, and FCC-mediated transactions. A project that demonstrably drives FAssets minting volume or Smart Account usage is, in effect, contributing to the FLR burn flywheel, which is a measurable form of ecosystem alignment.

3.5 Infrastructure gaps and developer pain points

INFERENCE Several infrastructure gaps are visible in the current ecosystem. First, there is no production-grade FCC orchestration layer: developers building FCC extensions must work directly with the scaffold and the InstructionSender pattern, which is workable but verbose. Second, there is no PMW SDK that abstracts the cosigner and data-provider relay flow for application developers; the primitive is exposed but not packaged. Third, the FDC's attestation latency is bounded by the round structure of the attestation protocol, which makes it suitable for near-real-time but not high-frequency use cases — a known constraint that affects product design. Fourth, there is no credible institutional compliance layer (KYC, AML, accreditation attestation) that integrates with Flare DeFi, despite the obvious institutional demand signal.

INFERENCE Developer pain points cluster around three areas: the learning curve for FCC (Go, TEE attestation, instruction relay), the operational burden of running FDC verifiers and indexers for external chains, and the absence of a unified SDK that combines FTSO, FDC, and FCC into a single developer experience. Each of these is itself a hackathon-worthy project, but each is also an infrastructure play rather than a product play — and infrastructure plays typically score lower on the 'product usefulness' criterion unless they are paired with a consuming application.

3.6 Commercial and institutional context

VERIFIED FACT Flare's recent commercial milestones include: Kraken's integration of USDT0 on Flare (April 2025), giving Flare access to the USD 140+ billion stablecoin liquidity pool; BitGo's regulated custody for FLR and SGB in the US and Europe (June 2025), enabling institutional access with full compliance; VivoPower's USD 100 million XRP treasury strategy partnership with Flare (June 2025); EasyA's developer onboarding series targeting XRPL builders (June 2025); and the Hyperliquid FXRP/USDH spot market.

INFERENCE Read together, these milestones define a clear institutional go-to-market motion: custody (BitGo) enables regulated holders; corporate treasuries (VivoPower) create demand for XRP-native yield; stablecoin liquidity (Kraken, USDT0) provides a counter-asset for trading pairs; and Hyperliquid provides a venue for FXRP pair trading. The missing piece is a product layer that ties these together for an institutional user — a treasury management protocol that can hold FXRP and sFLR, rebalance across venues, manage risk privately, and prove solvency to auditors. This is the gap Aegis targets.

3.7 Underserved users and unmet demand

INFERENCE The most underserved user segment in the Flare ecosystem is the institutional XRP or FLR holder — a corporate treasury (VivoPower and its imitators), a family office, a regulated fund, or a large individual holder — who wants to earn yield on Flare but cannot use the existing retail-oriented stack. This user has four needs that no current Flare product meets: (1) confidentiality — position visibility would move markets or violate internal policy; (2) verifiable solvency — auditors and limited partners must be able to verify without seeing positions; (3) cross-chain execution — the ability to move between Flare, XRPL, Base, and Hyperliquid without managing multiple wallets and bridges; and (4) autonomous risk management — the ability to set policy and have it enforced by an agent that does not require constant human supervision. Each of these maps directly onto a Flare primitive that is underutilised today.

4. Historical Winner Analysis

Flare has run two prior hackathons of direct relevance: the in-person Flare x Google Cloud Verifiable AI Hackathon at UC Berkeley (7–9 March 2025) and the virtual edition (7–14 March 2025). Together they distributed USD 100,000 in prizes across four tracks: AI x DeFi (DeFAI), RAG Knowledge, Social AI Agents, and Consensus Learning. The Summer Signal is the successor programme, narrower in scope (two bounties rather than four tracks, USD 12K rather than USD 100K) but strategically continuous with the Verifiable AI thesis.

4.1 The prior winners — what they built

4.1.1 2DeFi — in-person, AI x DeFi 1st and RAG 2nd

VERIFIED FACT 2DeFi, built by Alex Lu and Hitharth Khurana, is a Gemini-powered platform that helps TradFi users transition to DeFi on Flare through Google login, portfolio analysis, and guided investment strategies executed inside TEEs. Key features: Google Sign-In creates an embedded Flare wallet on signup; a custom DeFi strategy is generated after signup featuring staking, liquidity pools, swaps, and memes; the system integrates Sceptre for sFLR staking and Blazeswap for LPs and swaps; Gemini models scan a Robinhood portfolio from a single photograph to assess risk and assets; and an automated support agent answers real-time queries.

4.1.2 Flare Fact Checker — in-person, Social AI Agents 1st

VERIFIED FACT Built by Luka Skeledzija, Mark Znidar, and Gregor Krzmanc, Flare Fact Checker is an AI and RAG-based verification tool for scientific claims on Farcaster. Verifier nodes run inside TEEs to prevent tampering with the knowledge base and language models; each node returns a correctness score, a short explanation, and a list of sources. Multiple verifier nodes attempt to reach consensus, and rewards are distributed using Shapley values of each node's contribution.

4.1.3 Command Flare — in-person, RAG 1st and Consensus Learning 3rd

VERIFIED FACT Built by Ravi Riley, Rohan Patra, and Vardhan Shorewala at UC Berkeley, Command Flare is a stateful AI-driven interface that combines Graph RAG and Vector RAG in TEEs to enable conversational execution of swaps, lending, and borrowing. The Letta module retains conversational context for personalised, continuous user experiences, and UI components are generated through natural language commands.

4.1.4 NFT Deep Appraisal — in-person, Consensus Learning 1st

VERIFIED FACT Built by Roger Mas and Mohul Shukla, first-year UC Berkeley EECS students, NFT Deep Appraisal combines outputs from Gemini, Claude, and Llama inside Google Confidential Space TEEs with vTPM attestation. Confidence-based weights (a 70/30 price-consistency ratio) give higher influence to models with consistent and coherent reasoning, and the system queries the Moralis API for real-time data.

4.1.5 Quince Finance — virtual, AI x DeFi 1st

VERIFIED FACT Quince Finance introduced voice-controlled DeFi on Flare. Users sign in with Google to access a TEE-stored wallet and issue spoken commands such as 'Swap 50 FLR for SGB' to stake, swap, or manage liquidity hands-free. vTPM tokens provide authentication; TLS and dual Google ID verification protect user data.

4.1.6 ScribeChain — virtual, Social AI Agents 1st

**VERIFIED
FACT**

ScribeChain won the Social AI Agents track in the virtual edition. The Flare news summary describes it as an off-chain verifiable AI project that reshapes user experiences in social media; the full project description is available on the DoraHacks organisation page for Flare.

4.2 Recurring patterns across winners

INFERENCE

Six patterns recur across the prior winner set. **First**, every winner used a TEE for at least one load-bearing function — never as decoration. **Second**, every winner combined Flare primitives with an external AI model (Gemini, Claude, Llama) running inside the TEE, rather than calling the model from a standard backend. **Third**, every winner invested heavily in onboarding UX: Google Sign-In, voice commands, conversational interfaces, Farcaster integration, photograph-based portfolio input. The pattern is unmistakable: the prior hackathon privileged accessibility. **Fourth**, every winner integrated at least one existing Flare protocol or product (Sceptre, Blazeswap, FTSO). **Fifth**, the demo wow-moments were all transformations from an opaque input to a verifiable on-chain output. **Sixth**, the teams that won multiple prizes (2DeFi, Command Flare) had unusually broad technical surface area — they touched more of the stack than single-prize winners.

4.3 What these patterns imply for the Summer Signal

INFERENCE

The Summer Signal differs from the Verifiable AI Hackathon in two important respects. First, the tracks are bounties for product categories (interoperable assets, confidential compute) rather than AI sub-disciplines, so the AI component is now a means rather than an end. Second, the smaller prize pool and longer duration (six weeks versus one) select for fewer, more serious submissions rather than a high-volume demo fest. The implication is that the winning submission should be deeper rather than broader — a complete product with a real user, not a feature demonstration. The prior winners' emphasis on onboarding UX remains relevant, but the Summer Signal will reward depth of Flare integration over polish of the AI surface.

4.4 Reasons projects lost despite technical excellence

INFERENCE

Although we cannot access the full submission pool, several failure modes are inferred from the contrast between the winners' descriptions and the typical Verifiable AI Hackathon submission pattern. **First**, projects that used a TEE only to wrap an external LLM call (no attestation, no Flare data protocol consumption) likely scored poorly on Flare integration quality — the TEE was cosmetic. **Second**, projects that targeted an AI use case with no clear Flare-specific angle (e.g., a generic chatbot that happened to be deployed on Flare) likely scored poorly on product usefulness, because the chain choice was arbitrary. **Third**, projects with weak demos — no live on-chain interaction, only a slide walkthrough — likely scored poorly on technical execution regardless of code quality, because judges could not verify the work. **Fourth**, projects that did not clearly separate prior work from new work would have failed the 'evidence of new work' criterion, which the Summer Signal rubric makes explicit.

4.5 Comparable hackathons from competing ecosystems

**VERIFIED
FACT**

Comparable ecosystem hackathons include the Solana IGNITION hackathon (568 final projects, tracks in Web3, DeFi, Gaming, and NFTs, sponsored by Solana Ventures, Microsoft, Jump Crypto, Standard Chartered, Forte, and Metaplex), the Starknet Re{ignite} Hackathon (865 builders, 111 teams, 100+ submissions across Payments, Gaming, DeFi, and Consumer apps), and the broader DoraHacks catalog. A 13x hackathon winner interviewed by DoraHacks emphasised that what helps projects stand out to judges is strategic project selection — choosing what to build based on the specific organiser's priorities rather than generic appeal.

INFERENCE

The cross-ecosystem pattern is that grand-prize winners in sponsor-led hackathons consistently (a) use the sponsor's flagship technology in a load-bearing way, (b) target a use case the sponsor's business development team is already trying to monetise, and (c) demonstrate a path to real users. Solana IGNITION's grand prize winners were disproportionately in DeFi and infrastructure; Starknet Re{ignite}'s were disproportionately in payments and consumer apps that exercised Starknet's specific strengths. The analog for Flare is FCC and PMW in a treasury context.

5. Hidden Judging Psychology

The published rubric is the floor of what judges evaluate, not the ceiling. This section estimates the implicit factors that influence how judges actually score, and assigns each a rough weighting based on the evidence available. The weightings are estimates, not measurements.

5.1 Estimated implicit weighting

Implicit factor	Est. weight	Supporting evidence
Flare integration depth	25%	Only criterion framed as a binary test in the rubric; organiser's strategic priority.
Product usefulness	20%	Listed first in the rubric; emphasised in submission requirements (target user, traction).
Clarity & post-hack trajectory	15%	Rubric criterion; reinforced by the ask for roadmap and traction signals.
Technical execution & demo quality	15%	Rubric criterion; judges with technical depth will scrutinise attestation flows.
Evidence of new work	10%	Explicit rubric criterion; reinforced by the prior-work/new-work separation rule.
Strategic narrative alignment	8%	Implicit: does the project tell the Flare 2.0 story the organisers want to tell?
Polish and production readiness	4%	Implicit: working deployment on Coston2/Songbird/Mainnet vs. localhost only.
Team credibility signals	3%	Implicit: prior shipping history, communication quality, response to questions.

Weightings are Evidence-Based Inference. They are estimates intended to guide design priorities, not measurements of judge behaviour.

5.2 How judges implicitly evaluate ambition

INFERENCE Judges reward ambition that is matched by execution. A project that attempts something genuinely difficult — a working FCC extension with real attestation, a PMW-mediated cross-chain transaction, a verifiable AI agent that does useful work inside a TEE — and pulls it off will outscore a project that attempts something easy and executes it flawlessly. The reverse is also true: ambition without execution is actively penalised, because it signals that the team mis-scoped. The implication is that the right level of ambition is the highest level at which the team can credibly demo a working result. For a six-week build by a competent team, that level is: one FCC extension, one PMW-mediated cross-chain flow, one AI model inside the TEE, and a clean institutional dashboard.

5.3 Founder signals and team credibility

INFERENCE Judges form an impression of the team within the first thirty seconds of the demo. The signals they read — consciously or not — include whether the team articulates a thesis before showing features, whether they can answer technical questions about their own architecture without deflecting, whether they have thought about the post-hackathon path, and whether they have any external validation (a pilot conversation, a waitlist, a partner letter). The submission requirements explicitly invite traction signals, which means judges will be primed to look for them. A team that arrives with a one-paragraph letter of interest from a Flare ecosystem partner will outscore a team with better code but no external signal.

5.4 The 'meaningful Flare use' test

INFERENCE

The single most powerful implicit test a technical judge applies is the 'meaningful Flare use' test, which corresponds to the explicit 'Flare integration quality' criterion. The judge asks: 'If I removed Flare from this project, what would break?' If the answer is 'nothing significant' — if the project could run equally well on Ethereum, Base, or a conventional Web2 backend with a database — the integration is cosmetic and the project will be downscored heavily. If the answer is 'the entire product would cease to function' — because the product depends on FDC attestations that no other chain provides, or on PMW cross-chain execution that no other chain provides, or on FCC TEE attestation that no other chain provides — the integration is load-bearing and the project will be upscored. The design implication is absolute: every Flare primitive used must be load-bearing. If a primitive is not load-bearing, either remove it or redesign the product so that it is.

5.5 Emotional impact and the 'wow' moment

INFERENCE

Judges are human, and after reviewing dozens of submissions the projects that remain in memory are those that produced a visceral reaction. The wow moment for Flare is the confidentiality-to-verifiability transformation: a sensitive input goes into a TEE, and a verifiable proof comes out. A demo that dramatises this transformation — for example, by showing an auditor verifying solvency of a private vault without ever seeing the positions — produces an emotional impact that no feature list can match. The wow moment must be explicitly designed into the demo, not left to chance.

5.6 The judge's cognitive load

INFERENCE

Judges review many submissions in a constrained window. They will not read every line of every README. They will skim the project description, watch the demo video (or click the live demo if it loads in under five seconds), and spot-check the GitHub. The implication is brutal but useful: the submission's first paragraph, the demo's first fifteen seconds, and the GitHub's README header must collectively communicate the thesis, the Flare integration, and the wow moment. Anything buried below the fold is, for practical purposes, invisible to most judges.

6. Ecosystem Opportunity Matrix

This section ranks the opportunities visible in the Flare ecosystem across nine dimensions. The objective is to identify which opportunities justify a hackathon build and which are better suited to a longer-horizon grant or startup effort. Each opportunity is rated High, Medium, or Low on each dimension; the ratings are Evidence-Based Inference.

#	Opportunity	Urgency	Ecosystem impact	Technical feasibility	Commercial potential	Differentiation	Judge appeal	Sponsor appeal
1	Institutional XRP treasury management (FCC + PMW + FAssets)	H	H	M	H	H	H	H
2	Verifiable AI risk/credit underwriting engine in TEE	H	H	M	H	H	H	H
3	Private MEV-resistant DEX with TEE order matching	M	M	L	M	H	M	M
4	PMW-based cross-chain arbitrage / liquidity router	H	H	M	M	H	H	H
5	Compliance layer (KYC/AML/accreditation attestation) for Flare DeFiH	M	M	M	H	H	M	H
6	FDC-based proof-of-reserves for exchanges and custodians	M	H	H	M	M	M	H
7	Confidential AI agent marketplace (verifiable inference)M	M	M	M	M	H	H	M
8	RWA tokenization with AI pricing + FDC attestation	M	M	M	H	M	M	M
9	FCC orchestration SDK / dev tooling	H	H	H	L	M	L	M
10	Verifiable federated learning coordinator on Flare	L	M	L	M	H	M	L
11	XRPFi insurance / smart contract risk cover	M	M	M	M	M	M	M
12	AI-powered cross-chain portfolio rebalancer (retail)	M	L	H	M	L	M	L
13	FXRP-backed overcollateralized stablecoin	M	H	M	H	M	M	H
14	Private DAO voting with TEE tallying	L	L	H	L	M	L	L
15	FDC-based weather/crop insurance for emerging marketsM	M	M	H	M	H	M	L

Ratings: H = High, M = Medium, L = Low. Ratings are Evidence-Based Inference based on ecosystem gaps, sponsor priorities, technical complexity, and historical winner patterns.

6.1 Reading the matrix

INFERENC E

Three opportunities cluster at the top of the matrix across most dimensions: **institutional XRP treasury management** (#1), **verifiable AI risk/credit underwriting** (#2), and **PMW-based cross-chain arbitrage / liquidity routing** (#4). Each of these exercises multiple Flare primitives in a load-bearing way, addresses a real and current gap, and aligns with documented Flare commercial priorities (institutional onboarding, verifiable AI, cross-chain execution). The recommended project, Aegis, is a deliberate fusion of #1 and #2 with elements of #4 — it does not split the team's attention across opportunities but combines the three highest-leverage ones into a single coherent product.

INFERENC E

Opportunities that score high on urgency but lower on judge appeal — particularly **FCC orchestration SDK** (#9) — are valuable infrastructure but are poorly suited to a hackathon because their value is realised through downstream consumers rather than through a self-contained demo. They are stronger candidates for a Flare Foundation grant than for a six-week hackathon build. The same applies to **compliance layer** (#5), which is high-value but requires partnerships with regulated custodians to demonstrate real impact — a dependency that is hard to satisfy within the hackathon window.

6.2 Cross-cutting opportunities worth noting

INFERENCE

Three cross-cutting opportunities are worth flagging even though they are not the recommended primary build. First, **FDC-based proof-of-reserves** (#6) is a near-term institutional need that could be a strong secondary feature embedded in a larger product; Aegis incorporates it as a verifiable solvency component. Second, **FXRP-backed overcollateralized stablecoin** (#13) is a high-impact primitive that has no Flare-native implementation yet, but it requires meaningful liquidity to demonstrate, which makes it a weak standalone hackathon play. Third, **weather/crop insurance** (#15) is a canonical FDC use case that demos well but has lower commercial urgency for Flare's current institutional focus; it is a good fallback concept if institutional access proves infeasible.

7. Competitive Landscape Prediction

This section predicts what the majority of participants are likely to build, classifies the expected submissions by category, and explains how to outperform each category. The classifications are Reasoned Speculation based on prior Flare hackathon patterns, the track structure, and the publicly visible early BUIDL submissions.

7.1 Predicted submission categories

Category	Expected share	Saturation	Originality	Judge fatigue	Win probability
AI onboarding assistants (2DeFi/Quince clones)	~25%	Extremely common	Low	High	Low
Generic RAG chatbots over Flare docs	~15%	Extremely common	Low	High	Low
Simple FXRP wrappers (basic lending/staking/AMM)	~20%	Common	Low-Med	Med	Low
Voice/conversational DeFi interfaces	~8%	Common	Med	Med	Low-Med
NFT appraisal / consensus-learning demos	~7%	Common	Med	Med	Low-Med
FCC "Hello World" extensions (minimal TEE use)	~10%	Common	Low	High	Low
Private treasury / private vault (FCC)	~5%	Moderate	Med	Low	Med
Cross-chain PMW demos	~3%	Rare	High	Low	Med-High
Verifiable AI agents (production-grade)	~4%	Rare	High	Low	Med-High
Hybrid institutional product (FCC + PMW + FAssets)	<2%	Highly differentiated	Very high	Very low	High
Other / uncategorised	~1%	N/A	N/A	N/A	Low

Predictions are Reasoned Speculation. Share estimates assume approximately 80–120 final submissions from the ~415 registered hackers, based on typical DoraHacks conversion rates.

7.2 Early BUIDL signals observed

VERIFIED FACT

At least two BUIDLs were publicly visible at the time of research: **PrivyRoll Signal** (a private roll-call / attendance project using FCC, published 12 July 2026) and **BridgeSafe — Private Treasury** (a private treasury concept published in late July 2026). The presence of a 'private treasury' submission confirms that the FCC-based treasury category will be contested — but the early timing and the naming suggest a simpler formulation than the institutional-grade protocol proposed in this report.

INFERENCE

The strategic implication is that Aegis must differentiate sharply from BridgeSafe-style entries. The differentiation strategy has three components. First, **scope**: Aegis is a full treasury and risk layer, not a single-feature vault; it manages positions across multiple chains, not just balances. Second, **primitives used**: Aegis uses PMW for cross-chain execution and FDC for solvency attestation, in addition to FCC; a simpler treasury entry will use only FCC. Third, **commercial framing**: Aegis is explicitly positioned for the VivoPower-class institutional user, with a named business model and pilot outreach — a framing that no early BUIDL has yet claimed.

7.3 How to outperform each category

7.3.1 vs. AI onboarding assistants (2DeFi/Quince clones)

INFERENCE

These entries will be polished and accessible but will use Flare superficially — typically only FTSO for price display and an embedded wallet. Outperform by using FCC, PMW, FDC, and FAssets in combination. The 'meaningful Flare use' test will eliminate most of this category at the integration-quality criterion.

7.3.2 vs. Generic RAG chatbots



These entries will use TEEs for LLM inference but solve no real problem. Outperform by solving an institutional problem with measurable economic value, and by using FDC for verifiable external data rather than only RAG over static documents.

7.3.3 vs. Simple FXRP wrappers



These entries will use FAssets but not FCC, FDC, or PMW. Outperform by using FAssets as one component of a multi-primitive stack — the FXRP is the asset, but the FCC is the confidentiality, the FDC is the verification, and the PMW is the cross-chain execution.

7.3.4 vs. FCC Hello World extensions



These entries will deploy the scaffold FCC extension with minimal modification. Outperform by building a substantive FCC extension that does real work (AI risk inference, position computation, policy enforcement) and by combining it with the other primitives.

7.3.5 vs. Private treasury / private vault (FCC)



This is the most direct competitive category. Outperform on three axes: (a) use PMW for cross-chain execution, which simple treasury entries will not; (b) use FDC for verifiable solvency attestation, which simple treasury entries will not; (c) frame the product institutionally with named pilots and a credible business model, which simple treasury entries will not.

7.3.6 vs. Cross-chain PMW demos



These entries will demonstrate PMW but may lack a product wrapper. Outperform by embedding PMW inside a coherent product (treasury management) rather than presenting it as a standalone capability demo.

7.3.7 vs. Verifiable AI agent demos



These entries will run AI in TEEs but may lack institutional framing. Outperform by giving the AI a real institutional job (risk management, automated hedging, compliance attestation) rather than a generic chatbot or trading-agent role.

8. Grand Prize Strategy — Idea Discovery and Selection

This section documents the structured discovery process. We generated more than fifty candidate concepts, scored each across fourteen weighted criteria, and refined through multiple elimination rounds until only three remained. The full shortlist is presented, the eliminations are justified, and the winning concept is then developed in full in Section 9.

8.1 Scoring methodology

Each concept is scored on a 1–5 scale across fourteen criteria, with weights reflecting the implicit judging model from Section 5:

Criterion	Weight	What 5 means
Flare integration depth	15%	Uses FCC + PMW + FDC + FTSO + FAssets, all load-bearing
Product usefulness	12%	Solves a real, current, named pain point
Ecosystem alignment	10%	Advances Flare 2.0 thesis; matches sponsor priorities
Sponsor alignment	8%	Validates a Flare commercial partnership (BitGo, VivoPower, etc.)
Technical feasibility (6 weeks)	10%	Achievable by a competent 2–3 person team
Differentiation	8%	Unlikely to be replicated by another entrant
Business potential	7%	Clear revenue model and customer
User value	6%	Material improvement for the target user
Scalability	5%	Grows beyond the demo without redesign
Demo potential	8%	Produces a visceral wow moment live
Documentation potential	3%	Readable architecture and clear writeup
Open-source potential	3%	Reusable components for the ecosystem
Investment potential	3%	Fundable as a standalone company
Judge excitement	2%	Emotional impact on the panel

Final weighted score = sum of (score × weight) × 20, normalised to a 0–100 scale.

8.2 The fifty candidate concepts

The full set of fifty candidate concepts is listed below. Each is tagged with its primary bounty (I1 = Interoperable Asset Products; C2 = Confidential Compute Apps; B = both). Scores are the weighted totals out of 100; only the top fifteen are shown with full scores to keep the table readable; the remainder are listed for completeness with their approximate tier.

#	Concept	Track	Score
1	Aegis — Institutional XRP treasury + verifiable AI risk + PMW cross-chain	B	93
2	Verifiable AI credit/underwriting engine (FDC + FCC + AI in TEE)	B	88
3	PMW-based cross-chain liquidity router / arbitrage	B	84
4	Confidential AI risk-managed FXRP stablecoin issuer	B	82
5	FDC-based proof-of-reserves + solvency attestation service	I1	78
6	Compliance layer (KYC/AML attestation) for Flare DeFi	B	76
7	Verifiable AI agent marketplace (TEE-secured inference)	C2	74
8	Private MEV-resistant DEX with TEE order matching	B	72
9	RWA tokenization with AI pricing + FDC attestation	I1	70

10	AI-powered cross-chain portfolio rebalancer (retail)	B	68
11	FCC orchestration SDK / developer tooling	C2	66
12	XRPfi insurance / smart contract risk cover protocol	I1	64
13	FXRP-backed overcollateralized stablecoin	I1	63
14	Autonomous XRP market maker agent (TEE + PMW)	B	62
15	Cross-chain private vault (FXRP yield across Base/Hyperliquid)	B	61

Row 1 (Aegis) is highlighted as the recommended concept. Scores are Evidence-Based Inference.

8.2.1 Remaining concepts (tier D, approximate scores 40–60)

- 16. FDC-based weather/crop insurance for emerging markets (I1, ~59)
- 17. Private DAO voting with TEE tallying (C2, ~58)
- 18. Verifiable federated learning coordinator on Flare (C2, ~57)
- 19. FXRP options/perpetuals DEX (I1, ~56)
- 20. Self-repaying FXRP loans against stXRP collateral (I1, ~55)
- 21. FXRP restaking protocol (I1, ~54)
- 22. Cross-chain yield strip / principal tokens (I1, ~53)
- 23. TEE-based custodial wallet for institutions (C2, ~52)
- 24. Anti-fraud payment verification (FDC + AI in TEE) (B, ~52)
- 25. Verifiable AI bounty hunter agent (C2, ~51)
- 26. Confidential social reputation system (C2, ~50)
- 27. TEE-secured cross-chain bridge attestation (B, ~50)
- 28. Private medical records attestation (C2, ~49)
- 29. Verifiable inference for on-chain credit underwriting (B, ~49)
- 30. On-chain credit scoring with privacy (FDC + TEE) (C2, ~48)
- 31. Private intent-based DEX (orders matched in TEE) (B, ~48)
- 32. Decentralized XRP bridging optimizer (I1, ~47)
- 33. FXRP liquidity bootstrapping pool platform (I1, ~47)
- 34. XRP-native AMM with concentrated liquidity (I1, ~46)
- 35. Cross-chain yield aggregator via FXRP OFT (I1, ~46)
- 36. FXRP-based payment rails for merchants (I1, ~45)
- 37. XRP treasury management dashboard for corporates (I1, ~45)
- 38. Cross-chain BTC-XRP-FLR tri-collateral stablecoin (I1, ~44)
- 39. FXRP/FLR leveraged yield farming vault (I1, ~44)
- 40. XRP-backed flash loan protocol (I1, ~43)
- 41. Confidential portfolio manager (rebalance in TEE) (C2, ~43)
- 42. Verifiable LLM inference marketplace (C2, ~42)
- 43. Confidential AI-powered risk assessment for DeFi (C2, ~42)
- 44. Private KYC/attestation service (C2, ~41)
- 45. TEE-secured on-chain AI oracle (C2, ~41)
- 46. FXRP-collateralized stablecoin (overcollateralized) (I1, ~40)
- 47. Verifiable AI trading agent (autonomous, in TEE) (C2, ~40)
- 48. MEV-resistant private order execution (C2, ~40)
- 49. Cross-chain portfolio rebalancing agent (B, ~40)

- 50. XRPFi insurance protocol (smart contract risk) (I1, ~40)

8.3 Elimination rounds

8.3.1 Round 1 — Eliminate concepts scoring below 60

Concepts 16–50 (35 concepts) are eliminated. The common failure modes are: (a) insufficient Flare primitive depth (e.g., concepts 32–34 use only FAssets); (b) weak demo wow-moment (e.g., concepts 17, 27, 41); (c) narrow commercial framing (e.g., concepts 18, 25, 28); (d) excessive implementation complexity for a six-week window (e.g., concepts 8, 19, 30, 31); or (e) direct overlap with stronger concepts above (e.g., concept 47 is a subset of concept 1).

8.3.2 Round 2 — Eliminate concepts scoring 60–75

Concepts 6–15 (10 concepts) are eliminated. The common failure modes are: (a) infrastructure rather than product framing (concepts 11, 12 — stronger as grants than as hackathon builds); (b) dependence on liquidity or partner access that cannot be demonstrated in six weeks (concepts 8, 13); (c) redundancy with a higher-scoring concept (concepts 4, 14, 15 are subsets of concept 1). The remaining top five are concepts 1–5.

8.3.3 Round 3 — Top 5 to Top 3

Concept 5 (FDC-based proof-of-reserves) is eliminated because, although it scores well on ecosystem importance and sponsor appeal, it is an infrastructure service rather than a complete product — it would need to be embedded in a consuming application to demonstrate value. Concept 4 (FXRP stablecoin issuer) is eliminated because it requires meaningful liquidity to demo credibly and because the verifiable AI component is decorative rather than load-bearing. The remaining top three are concepts 1, 2, and 3.

8.4 The Top 3

Rank	Concept	Score	Strengths	Weaknesses
1	Aegis — Institutional XRP treasury	93	Verifiable AI risk in PMW; institutional framing; clear business model; strong PMW moment	Implementation complexity
2	Verifiable AI credit/underwriting engine (FDC + Str56G AI Arbitrator)	86	Verifiable AI risk in PMW; real underwriting problem; uses FDC; strong PMW moment	Need to use FDC; harder to demo today; harder to integrate with Flare today
3	PMW-based cross-chain liquidity router (4 arbitrage)	84	Page PMW showcase; clear technical depth; liquidity depth; arbitrage framing	Liquidity depth; arbitrage framing may feel extractive

8.5 Final selection

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Concept 1, **Aegis**, is selected as the recommended build. The decisive factors are: (a) it is the only concept that uses all five Flare primitives in a load-bearing way, maximising the dominant 'Flare integration quality' criterion; (b) it has the clearest institutional customer — the VivoPower-class XRP treasury — which is a verified commercial priority for Flare; (c) it has the strongest demo wow-moment (the confidentiality-to-verifiability transformation); and (d) it is the only concept that is competitive in both bounties simultaneously, maximising expected prize value. The implementation complexity is the principal risk, and the engineering roadmap in Section 9 is designed to mitigate it through aggressive scoping of the MVP.

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Concepts 2 and 3 are retained as fallback options. If PMW proves infeasible to integrate within the build window (a non-trivial risk given how new the primitive is), Concept 2 can be built with FCC + FDC + FAssets alone and remain highly competitive. If the institutional framing proves difficult to validate, Concept 3 can be repositioned as a developer-facing PMW showcase and still score well on technical execution.

9. The Winning Project — Complete Blueprint

9.1 Product Strategy

9.1.1 Project title

Aegis — A Verifiable, Confidential, AI-Managed Cross-Chain Treasury and Autonomous Risk Layer for XRP-Native Institutions on Flare.

9.1.2 Executive summary

Aegis is an institutional-grade treasury protocol that lets corporate XRP and FLR holders earn yield, manage risk, and prove solvency — all without exposing their positions to the public chain, to a custodial operator, or to the AI agent that manages their portfolio. Deposits are received as FXRP via Flare Smart Accounts or as sFLR via Sceptre-style liquid staking. Position state is computed inside Flare Confidential Compute (FCC) TEEs, which publish only a Merkle root of solvency on-chain. An AI risk agent, also inside a TEE, continuously monitors FTSO price feeds and FDC-attested cross-chain state, and executes rebalances or hedges via Protocol Managed Wallets (PMW) across Flare, XRPL, Base, and Hyperliquid. External auditors can verify solvency cryptographically without ever seeing individual positions.

9.1.3 Vision

To become the default treasury infrastructure for XRP-native institutions on Flare — the layer that converts passive XRP and FLR holdings into verifiably-managed, privately-executed, cross-chain yield positions, and that makes Flare the only credible settlement layer for institutional XRPFi.

9.1.4 Mission

Make it possible for any corporate treasury, family office, or regulated fund to hold XRP or FLR and earn Flare DeFi yield with the confidentiality of a prime broker, the verifiability of a public blockchain, and the autonomy of an AI risk officer — without trusting a single operator.

9.1.5 Problem statement

Corporate XRP treasuries are emerging as a recognised category. VivoPower committed USD 100 million to an XRP treasury strategy in partnership with Flare in June 2025. Other corporates are publicly considering similar allocations. These holders face four problems that no current Flare product solves. First, **confidentiality**: publishing treasury positions on a public chain would move markets, violate internal policy, and create competitive exposure. Second, **verifiable solvency**: auditors, boards, and limited partners require proof that the treasury is solvent — but they must be able to verify this without seeing individual positions. Third, **cross-chain execution**: yield opportunities exist across Flare (Sceptre, Blazeswap, Kinetic, Clearstar), XRPL (native liquidity), Base (via FXRP OFT), and Hyperliquid (FXRP/USDH) — but moving between them today requires multiple wallets, bridges, and gas tokens. Fourth, **autonomous risk management**: treasury operations require 24/7 risk monitoring and automated hedging that no off-the-shelf tool provides.

9.1.6 Target users

The primary ideal customer profile (ICP) is a corporate treasury, family office, or regulated fund holding USD 5 million or more in XRP or FLR and seeking yield on Flare. Secondary ICPs include: (a) XRP whales (individual holders above USD 1 million) seeking private yield; (b) DAOs and protocol treasuries holding FLR that want confidential rebalancing; (c) regulated custodians (such as BitGo) seeking a Flare-native yield product to offer their institutional clients.

9.1.7 User personas

Persona	Profile	Pain	Aegis solution
Maya — CFO of a public XRP treasury	USD 50M in XRP. Reports no published positions.	Can't easily compute position, quarterly	Aegis dashboard computes position, quarterly
David — CIO of a crypto-native family office	USD 200M across multiple tokens	Needs rebalancing, access to yield	Aegis Hyperliquid, spends balance, PMW
Sarah — Head of product at a regulated custody firm	Has clients who want yield	Can't hold Aegis vault in DeFi	Partner with BitGo custody

9.1.8 User journeys

Maya (corporate treasury CFO) onboards

Maya's team contacts Aegis. The Aegis onboarding contract mints an institutional vault on Flare, parameterised with Maya's risk policy (max drawdown 15%, max single-asset exposure 40%, hedge threshold at 10% drawdown). Maya's XRPL wallet — already in use at her corporate treasury — is mapped to a Flare Smart Account. She signs a single XRPL transaction that mints FXRP and deposits it into the Aegis vault. The FXRP deposit is attested via FDC and the vault balance is recorded. Maya's position is now live.

David (family office CIO) sets a risk policy

David logs into the Aegis dashboard (authenticated via his existing wallet). He sets a risk policy: target allocation 60% sFLR / 30% FXRP-yield-vault / 10% USDT0; rebalance band $\pm 5\%$; hedge at 8% drawdown by opening a short on Hyperliquid. The policy is committed to the Aegis Policy smart contract. The AI risk agent, running in a TEE, reads the policy and begins monitoring. When FTSO detects an 8% drawdown, the agent issues a rebalance instruction via PMW: it moves FXRP from the vault to Hyperliquid, opens the short, and returns the hedge receipt to the vault. The entire flow is verifiable on-chain; the position sizes are not.

Maya's auditor verifies solvency

At quarter-end, Maya's auditor (a Big Four firm) is granted read access to the Aegis solvency proof. The auditor queries the Aegis Verifier contract, which requests a fresh attestation from the FCC extension. The TEE computes the current vault state (assets, liabilities, hedge positions) and returns a cryptographic proof that assets exceed liabilities by a stated margin, without revealing the underlying amounts. The auditor verifies the proof on-chain and signs off. The auditor never sees Maya's positions; the board is satisfied; the regulator is satisfied.

9.1.9 Market opportunity

VERIFIED FACT VivoPower's USD 100 million XRP treasury commitment in June 2025 is a single verified data point, but it is a meaningful one: it demonstrates that the category of corporate XRP treasury exists and is allocating. The broader stablecoin and treasury management market — across all assets — is in the tens of billions of dollars; the XRP-specific subset is materially smaller today but is growing as XRP regulatory clarity improves (the US court ruling classifying XRP as a non-security, XRP-backed ETF inflows of approximately USD 1.5 billion, and the XRPL 3.2.1 upgrade all contribute to this trajectory).

INFERENCE If Aegis captures even a small percentage of a growing XRP treasury market, the protocol's assets under management could reach hundreds of millions of dollars within two to three years. At a 50 basis point management fee and a 10% performance carry, this corresponds to a seven-to-eight-figure annual revenue line, which is a fundable startup outcome. The hackathon does not need to deliver this scale; it needs to demonstrate the product that makes this scale plausible.

9.1.10 Existing solutions and why they fail

Existing solution	What it does	Why it fails for institutions
Sceptre (sFLR)	Liquid staking of FLR; accrues staking rewards.	Not Flare-aligned; no confidentiality; no AI risk management; no cross-chain execution.
Blazeswap / Enosys (AMM)	Swap and LP for sFLR pairs.	Public positions; no institutional compliance; no risk policy automation.
Kinetic (lending)	Borrow against FLR collateral.	Public positions; no confidentiality; no cross-chain collateral.
Firelight (stXRP)	Liquid staking of XRP via FAssets.	Single-strategy; no risk management; no institutional interface.
Clearstar (XRP yield vault)	Yield vault for FXRP.	Single-strategy; public; no AI risk layer; no cross-chain execution.
Custodial prime brokers (off-chain)	Confidential treasury management.	Trusted operator; not verifiable; not on-chain; not Flare-aligned.
2DeFi (prior hackathon winner)	AI onboarding to Flare DeFi.	Retail-focused; no confidentiality; no PMW; demo-grade, not production.

9.1.11 Unique insight



The unique insight that powers Aegis is that **Flare's three newest primitives — FCC, PMW, and FDC — are not three separate features but three components of a single institutional product.** FCC provides the confidentiality that institutions require; PMW provides the cross-chain execution that treasury management demands; FDC provides the verifiable attestation that auditors and regulators need. No other chain offers this combination. The product is not possible on Ethereum, Base, Solana, or any other EVM chain, because none of them enshrine all three primitives. This is the structural answer to the 'meaningful Flare use' test: remove Flare, and the product cannot exist.

9.1.12 Core innovation

The core innovation is the **confidential position computation with verifiable solvency** pattern. Position state is computed inside a TEE and never published in plaintext; only a Merkle root of the solvency proof is published on-chain. Any authorised verifier (auditor, regulator, board member) can request a fresh attestation and verify solvency cryptographically, without ever seeing the underlying positions. This resolves the apparent paradox between confidentiality and verifiability that has blocked institutional DeFi adoption on every other chain.

9.1.13 Value proposition

For the institutional user: **earn Flare DeFi yield on XRP and FLR with the confidentiality of a prime broker, the verifiability of a public blockchain, and the autonomy of an AI risk officer — without trusting a single operator.**
 For the Flare ecosystem: **the first reference implementation of FCC + PMW + FDC in combination, targeted at the institutional segment Flare's commercial team is actively courting.**

9.1.14 Competitive moat



The moat is compound. A competitor would need to replicate the FCC extension, the PMW integration, the FDC attestation flow, the AI risk model, the institutional dashboard, and — most importantly — the institutional trust and partner relationships. Each is individually replicable; the combination is not. Aegis's first-mover advantage in the institutional XRPFi category, if executed competently, creates a network effect: institutions prefer the treasury protocol that other institutions already use, because shared auditors and shared compliance frameworks reduce onboarding friction.

9.1.15 Long-term vision

Beyond treasury management, Aegis becomes the confidential execution layer for institutional DeFi on Flare: a generalised pattern in which any institutional workflow that requires confidentiality, verifiable solvency, cross-chain execution, and autonomous risk management can be expressed as an Aegis vault. Applications include regulated lending, tokenised fund administration, cross-border treasury, and — eventually — central bank digital currency interoperability. The vault abstraction is the long-term platform.

9.1.16 Success metrics (KPIs)

Metric	Hackathon target (demo)	6-month target	12-month target
Total value deposited in Aegis vaults	USD 50K test funds on Costello	USD 5M (pilot)	USD 50M+
Number of institutional pilots in conversation	2 LOIs secured	5 pilots live	15+ institutions
Number of Flare primitives used load-bearing	5 (FTSO, FDC, FCC, PMW, F Assets)	5 Assets	5+ (add Smart Accounts, FTSO v2)
FCC extensions deployed	1 production extension	3 (vault, risk, compliance)	6+
Auditor verifications completed	1 demo verification	10 quarterly verifications	50+ verifications/quarter
GitHub stars on the open-source vault contracts	50+	500+	2,000+
Protocol revenue (annualised)	0 (pre-revenue)	USD 100K run rate	USD 1M+ run rate

9.2 Research and Innovation

9.2.1 Novel research contribution

Aegis contributes a verifiable-confidential position computation pattern: a method by which an institutional treasury's position state (assets, liabilities, hedges, P&L;) can be computed inside a TEE and attested to external verifiers through a Merkle-root proof of solvency, without ever publishing the underlying positions in plaintext. The pattern is general: it applies to any workflow that requires (a) confidentiality against public observation, (b) verifiable solvency to authorised auditors, and (c) on-chain enforcement of risk policy. The specific contribution is the composition of FCC attestation, FDC external-state verification, and PMW cross-chain execution into a single institutional product — a composition that has not, to our knowledge, been demonstrated before.

9.2.2 Technical innovation

Four technical innovations are present. First, the **Policy-as-Code vault**: risk policies (drawdown limits, exposure limits, hedge triggers) are expressed as on-chain smart contract parameters that the TEE-bound AI agent must respect. Second, the **verifiable solvency attestation**: a TEE-produced Merkle root of the vault's solvency state, published on-chain and verifiable by any auditor with a cryptographic proof. Third, the **PMW-mediated autonomous rebalance**: an AI agent inside a TEE issuing cross-chain execution instructions through PMW, with the data-provider consensus providing the safety wrapper. Fourth, the **AI risk model inside the TEE**: a model that consumes FTSO and FDC data, computes risk metrics, and emits policy actions — all verifiable through TEE attestation.

9.2.3 Scientific contribution (if applicable)

The work does not claim a novel ML architecture; the AI risk model uses established techniques (gradient-boosted risk scoring, simple volatility forecasting, configurable policy thresholds). The scientific contribution is at the systems level: it demonstrates that the Flare 2.0 stack (FCC + PMW + FDC) can support a class of institutional applications that have previously been impossible on-chain, and it provides a reference implementation that future research and production work can extend.

9.2.4 State-of-the-art comparison

System	Confidentiality	Verifiable solvency	Cross-chain execution	Autonomous AI risk
Aegis (Flare)	Yes (FCC TEE)	Yes (FDC attestation)	Yes (PMW)	Yes (in-TEE agent)
Off-chain prime brokers	Yes (operational)	No (trust-based)	Manual	Limited
On-chain lending (Aave, etc.)	No (public)	Implicit (public)	No	No
Privacy chains (Aztec, etc.)	Yes (zk)	Partial	Limited	No
Verifiable AI projects (Flare private)	Partial (TEE)	No	No	Partial (LLM call)

9.2.5 Key assumptions

- FCC on Coston2 is sufficiently stable to support a non-trivial TEE extension for the duration of the build and judging window.
- PMW on Songbird (or Coston2) supports the XRPL cross-chain flows needed for the demo, including the ability to issue and verify XRPL transactions from an on-chain Flare protocol.
- FTSO price feeds for FXRP/FLR/USD and other relevant pairs are available and sufficiently fresh for risk monitoring.
- FDC supports the attestation types needed (XRPPayment, EVMTransaction, AddressValidity) and the latency is acceptable for the demo flow.
- The judging panel includes at least one technical judge capable of understanding the FCC attestation flow and the PMW execution path.

9.2.6 Design trade-offs

The principal design trade-off is **scope vs. depth**. Aegis could be broader (more chains, more strategies, more asset types) or deeper (more rigorous risk model, more complete compliance layer, more auditor tooling). We choose depth in the institutional treasury vertical over breadth across verticals, because the judging model rewards meaningful Flare integration over feature count. The second trade-off is **real PMW vs. simulated PMW**: if PMW integration proves infeasible within the build window, the demo can fall back to a simulated cross-chain flow with explicit labelling — but this would weaken the 'technical execution' score. The third trade-off is **institutional pilot vs. self-serve demo**: securing a real institutional letter of interest is high-value but uncertain; the demo must work without it.

9.2.7 Risks and limitations

- **Technical risk**: FCC and PMW are new; integration bugs and documentation gaps are likely. Mitigation: build the FCC scaffold extension early and validate the PMW flow on Coston2 in week one.
- **Scope risk**: the project is ambitious for six weeks. Mitigation: aggressive MVP scoping in Section 9.7; daily standups; cut features ruthlessly if behind.
- **Demo risk**: a live TEE demo can fail in front of judges. Mitigation: pre-recorded fallback video; demo contingency plan in Section 11.
- **Adoption risk**: no real institutional pilot may materialise. Mitigation: the demo works with test funds; pilot outreach is a bonus, not a dependency.
- **Regulatory risk**: institutional treasury management is a regulated activity in most jurisdictions. Mitigation: Aegis is positioned as infrastructure, not as a regulated entity; partner with regulated custodians (BitGo) for the regulated surface.

9.2.8 Future research directions

- Generalising the verifiable-confidential position pattern beyond treasury to lending, fund administration, and CBDC interoperability.
- Multi-model AI risk consensus (borrowing the NFT Deep Appraisal pattern) to reduce single-model bias in risk decisions.
- Federated learning across Aegis vaults for shared risk intelligence without position disclosure.
- Formal verification of the solvency proof circuit.
- Integration with regulated custodians (BitGo) for the institutional compliance surface.

9.3 Solution Architecture

9.3.1 Overall system architecture

Aegis is structured as a five-layer system. **Layer 1 (Asset Onboarding)** uses Flare Smart Accounts and FAssets to onboard XRP and FLR into Aegis vaults. **Layer 2 (On-Chain Vault Contracts)** holds the deposited assets and enforces the risk policy parameters. **Layer 3 (Confidential Compute Layer)** runs the FCC extension that computes position state and the AI risk model inside TEEs. **Layer 4 (Cross-Chain Execution)** uses PMW to execute rebalances and hedges across Flare, XRPL, Base, and Hyperliquid. **Layer 5 (Verification & Audit)** uses FDC to attest external chain state and to publish solvency proofs verifiable by external auditors. A frontend dashboard provides the institutional user interface; a developer SDK exposes the vault primitive to other builders.

```

+-----+
| LAYER 5 – VERIFICATION & AUDIT |
| FDC attestation | Solvency proof oracle | Auditor dashboard |
+-----+
| attests
+-----+
| LAYER 4 – CROSS-CHAIN EXECUTION (PMW) |
| XRPL wallet ops | Base OFT transfers | Hyperliquid hedging |
+-----+
| executes
+-----+
| LAYER 3 – CONFIDENTIAL COMPUTE (FCC) |
| Position computer | AI risk agent | Policy enforcement |
| (all inside TEEs, attested via FCC) |
+-----+
| reads/writes
+-----+
| LAYER 2 – ON-CHAIN VAULT CONTRACTS (Flare C-Chain) |
| VaultCore | PolicyRegistry | SolvencyRoot | InstructionSender |
+-----+
| deposits/withdraws
+-----+
| LAYER 1 – ASSET ONBOARDING |
| Flare Smart Accounts | FAssets (FXRP) | sFLR (Sceptre) |
+-----+
| user interacts
+-----+
| INSTITUTIONAL DASHBOARD + DEVELOPER SDK |
+-----+

```

Figure 1: Aegis five-layer system architecture (text rendering).

9.3.2 Solution architecture — request/response flows

Three primary flows traverse the layers. **Flow A (Deposit)**: the user signs an XRPL transaction via Flare Smart Accounts, which mints FXRP and deposits it into the VaultCore contract on Flare. The deposit is attested by FDC (XRPPayment attestation); the TEE position computer is notified via an InstructionSender event; the position state is updated inside the TEE; a new solvency root is published on-chain. **Flow B (Risk Rebalance)**: the AI risk agent inside the TEE detects a policy threshold breach (via FTSO price feed), issues a rebalance instruction to PMW, PMW signs and executes the cross-chain transaction (e.g., moves FXRP to Hyperliquid via OFT, opens a short), FDC attests the resulting Hyperliquid state, the TEE updates the position state, a new solvency root is published. **Flow C (Audit Verification)**: an auditor requests a solvency attestation via the Verifier contract, the TEE computes a fresh solvency proof, the proof is published on-chain, the auditor verifies it cryptographically without seeing the underlying positions.

9.3.3 AI / ML architecture

The AI risk model is a composite of three components running inside the TEE. **Component 1 (Risk Scorer)**: a gradient-boosted model (XGBoost or LightGBM) that ingests FTSO price feeds, FDC-attested cross-chain state, and on-chain vault parameters, and outputs a risk score (0–100) and a set of recommended actions. **Component 2 (Policy Engine)**: a deterministic rule engine that maps the risk score and current positions to specific policy actions (rebalance, hedge, deleverage) within the constraints set by the on-chain PolicyRegistry. **Component 3 (Action Executor)**: the module that translates policy actions into PMW instructions and submits them via the InstructionSender. The model is intentionally simple — the innovation is in the verifiable-confidential execution pattern, not in the model architecture.

9.3.4 Agentic AI architecture

The agent operates as a single-threaded loop inside the TEE: observe (read FTSO + FDC + vault state) → score (run Risk Scorer) → decide (run Policy Engine) → act (submit PMW instruction via InstructionSender if policy action is non-null) → attest (publish new solvency root) → sleep (until next FTSO round or threshold event). The loop is stateless across iterations except for the TEE-internal position state, which is rebuilt from on-chain attestations on each iteration to ensure the agent cannot drift from consensus state. This is a deliberate design choice: it makes the agent's behaviour verifiable from on-chain data alone.

9.3.5 Data architecture

On-chain data: vault parameters (PolicyRegistry), vault asset balances (VaultCore), solvency roots (SolvencyRoot contract), instruction log (InstructionSender). Off-chain data (TEE-internal): full position state (per-vault asset/liability/hedge breakdown), AI model weights, recent FTSO price history (cached), FDC attestation cache. External data sources: FTSO price feeds, FDC attestation API, Hyperliquid API (for hedge execution state), Base RPC (for OFT transfer state). The TEE-internal state is the only confidential data; everything else is either on-chain or reproducible from on-chain data.

9.3.6 Cloud architecture

Aegis is primarily on-chain and TEE-based, with minimal cloud dependency. The cloud surface is limited to: (a) the institutional dashboard frontend (deployed on Vercel or equivalent, static + edge functions); (b) the FDC verifier and indexer for external chains (run by Flare infrastructure providers, not by Aegis); (c) optional TEE attestation verification service for auditors who prefer a hosted interface. The architecture is deliberately cloud-minimal to maximise the 'meaningful Flare use' score: the product must not depend on a centralised backend for its core function.

9.3.7 Confidential compute / TEE architecture

The FCC extension is a Go HTTP server deployed inside a TEE (per the Flare Developer Hub quickstart). It exposes a POST /action endpoint that receives instructions from the InstructionSender contract via the FCC relay infrastructure, processes them (computing position state, running the AI risk model, emitting policy actions), and returns results that callers poll from the extension proxy. TEE attestation is verified on-chain during machine registration, so TEE signatures serve as proof of correct execution. The extension runs three logical modules: PositionComputer, RiskAgent, and SolvencyAttestor.

9.3.8 Cross-chain execution architecture (PMW)

PMW is the cross-chain execution primitive. The Aegis VaultCore contract is the protocol on Flare that owns PMW-managed wallets on XRPL, Base, and (via OFT) Hyperliquid. Keys are generated inside TEEs and never leave; they sign only when a weighted majority of data providers relays the instruction. Cosigners can be added for additional safety. For the demo, the primary PMW flow is XRPL: the vault moves FXRP between Flare and XRPL, or issues XRPL payments to settlement counterparties. Base and Hyperliquid flows use the FXRP OFT route for transfers and Hyperliquid API for hedging, attested via FDC.

9.3.9 Identity & access management

Vault ownership is bound to the depositor's XRPL address via Flare Smart Accounts. The XRPL keys govern the associated Flare Smart Account, so control stays anchored to the user's XRPL account. Auditor access is granted via a separate Verifier role on the SolvencyRoot contract; the auditor can request attestations but cannot see positions or move funds. Operator (Aegis team) access is limited to emergency pause and parameter updates, governed by a timelocked multisig.

9.3.10 Privacy architecture

Privacy is enforced by TEE confidentiality: position state is computed inside the enclave and never published in plaintext. The only on-chain outputs are asset movements (which are inherent to the vault's operation) and the solvency Merkle root (which is a hash, not the underlying data). Auditor attestations reveal only the boolean solvency result and aggregate metrics (total assets, total liabilities, surplus) — never individual position sizes. Optional zero-knowledge proofs (future work) could hide even the aggregate metrics, but the TEE-based approach is sufficient for the institutional use case and is far simpler to implement in six weeks.

9.3.11 Security architecture

Security rests on three layers. **Layer 1 (Smart Contract Security):** vault contracts are audited (internal review during the build, formal audit post-hackathon), use OpenZeppelin libraries, and are deployed behind a proxy with a timelocked multisig for upgrades. **Layer 2 (TEE Security):** the FCC extension runs inside an attested TEE; TEE identity is verified on-chain during registration; the extension's behaviour is deterministic given its inputs, so any deviation is detectable. **Layer 3 (Economic Security):** the vault is overcollateralised (assets always exceed liabilities), PMW requires data-provider consensus for any cross-chain movement, and the AI agent operates within strict policy limits that it cannot exceed.

9.3.12 Disaster recovery and business continuity

If the TEE fails or becomes unavailable, the vault enters a safe state: no new positions are taken, no rebalances occur, and the user can withdraw their deposited assets via an emergency exit path that does not depend on the TEE. If the AI agent emits an erroneous instruction, the Policy Engine's deterministic constraints prevent the instruction from violating the on-chain policy parameters. If PMW is unavailable, cross-chain execution pauses but on-chain Flare operations continue. The system is designed to fail safe rather than fail fast.

9.4 Technical Design

9.4.1 Component diagram (logical)

```

INSTITUTIONAL DASHBOARD (React/Next.js, hosted)
|
|--- User wallet (XRPL via Xaman / Ledger / Joey)
|--- Auditor verifier view (read-only)
|
v
[ Flare Smart Accounts ] -- maps XRPL addr -> Flare Smart Account
|
v
[ VaultCore.sol ] (Flare C-Chain)
|--- depositFXRP(asset, amount, policyId)
|--- withdraw(amount)
|--- emergencyExit()
|--- getBalance(user)
|
+---> [ PolicyRegistry.sol ] (risk params: maxDD, maxExposure, hedgeThreshold)
+---> [ SolvencyRoot.sol ] (stores Merkle root of solvency proof)
+---> [ InstructionSender.sol ] (relays instructions to FCC extension)
+---> [ VerifierRole.sol ] (auditor access control)
|
v
[ FCC EXTENSION (Go, inside TEE) ]
|--- PositionComputer (rebuilds state from on-chain events + FDC attests)
|--- RiskAgent (XGBoost model + Policy Engine)
|--- SolvencyAttestor (Merkle root publisher)
|--- ActionExecutor (issues PMW instructions)
|
v
[ PMW Layer ] -- controls wallets on:
|--- XRPL (settle FXRP, issue payments)
|--- Base (FXRP OFT transfers)
|--- Hyperliquid (open/close hedges)
|
v
[ FDC Attestations ] -- verifies external state:
|--- XRPPayment (XRPL transfers settled)
|--- EVMTransaction (Base OFT confirmed)
|--- AddressValidity (counterparty KYC)

```

9.4.2 Sequence diagram — risk rebalance flow

```

sequenceDiagram
    participant RiskAgent as RiskAgent(TEE)
    participant FTSS as FTSS
    participant VaultCore as VaultCore
    participant InstructionSender as InstructionSender
    participant PMW as PMW
    participant XRPL as XRPL
    participant FDC as FDC
    participant SolvencyRoot as SolvencyRoot

    FTSS->>RiskAgent: price
    RiskAgent->>FDC: [risk score computed; threshold breach detected]
    FDC->>RiskAgent: 
    RiskAgent->>PMW: --propose action (move FXRP to XRPL)----->
    PMW->>RiskAgent: 
    RiskAgent->>FDC: --policy check (on-chain)--
    FDC->>RiskAgent: --policy OK----->
    RiskAgent->>PMW: --instruction->
    PMW->>RiskAgent: --consensus (data providers)
    RiskAgent->>PMW: --sign & submit->
    PMW->>RiskAgent: --tx exec-->
    RiskAgent->>FDC: --receipt---
    FDC->>RiskAgent: --attest->
    RiskAgent->>FDC: --proof---
    FDC->>RiskAgent: --attestation result--
    RiskAgent->>SolvencyRoot: [TEE updates internal position state]
    SolvencyRoot->>RiskAgent: 
    RiskAgent->>XRPL: --publish new solvency root----->
    XRPL->>RiskAgent: 
  
```

Figure 2: Sequence diagram for an autonomous risk rebalance via PMW.

9.4.3 Data flow diagram

Inbound data flows: (1) FTSS price feeds → RiskAgent (TEE); (2) FDC attestation responses → PositionComputer (TEE); (3) on-chain vault events (deposits, withdrawals, policy changes) → PositionComputer (TEE) via the FCC relay. Outbound data flows: (1) PMW instructions → external chains (XRPL, Base, Hyperliquid); (2) solvency Merkle root → SolvencyRoot contract; (3) auditor attestation responses → Verifier contract. The TEE is the sole producer of solvency proofs and the sole issuer of PMW instructions; this is the security-critical data-flow invariant.

9.4.4 Database design

There is no traditional database. On-chain state (Flare C-Chain) is the canonical store for vault balances, policy parameters, and solvency roots. TEE-internal state (in-memory, rebuilt from on-chain data on each loop iteration) holds the confidential position breakdown and the AI model weights. Off-chain indexes (Hyperliquid API, Base RPC, XRPL) provide external state that is verified through FDC attestations rather than trusted directly. The architecture deliberately avoids a centralised database to preserve the 'meaningful Flare use' property: any state that lived in a centralised DB could be lost or tampered with, breaking the verifiability invariant.

9.4.5 API specification (core)

```
// On-chain (Flare C-Chain, Solidity) interface IVaultCore { function depositFXRP(uint256
amount, uint256 policyId) external; function withdraw(uint256 amount) external; function
emergencyExit() external; function balanceOf(address user) external view returns
(uint256); function policyOf(address user) external view returns (Policy memory); }
interface IPolicyRegistry { struct Policy { uint256 maxDrawdownBps; // e.g., 1500 = 15%
uint256 maxSingleExposureBps; // e.g., 4000 = 40% uint256 hedgeThresholdBps; // e.g., 800
= 8% address[] allowedAssets; // whitelist } function setPolicy(uint256 policyId, Policy
calldata p) external; function getPolicy(uint256 policyId) external view returns (Policy
memory); } interface ISolvencyRoot { function publishRoot(bytes32 root, uint256
surplusBps) external; // only TEE function verifySolveny(bytes32[] calldata proof,
bytes32 leaf) external view returns (bool); } interface IInstructionSender { function
sendInstruction(bytes calldata payload) external; // vault -> TEE function
getResponse(bytes32 instructionId) external view returns (bytes memory); } // Off-chain
(FCC extension, Go, inside TEE) // POST /action (called by FCC relay from
InstructionSender events) // Request: { instruction_id, payload, attestation_context } //
Response: { result, new_solveny_root, pmw_instructions[], fdc_requests[] }
```

9.4.6 SDK design

Aegis ships an open-source TypeScript SDK (@aegis/sdk) that exposes three modules: **VaultClient** (deposit, withdraw, query balance), **PolicyClient** (set and inspect risk policies), and **AuditClient** (request and verify solveny attestations). The SDK abstracts Flare RPC, FDC attestation retrieval, and TEE proof verification behind a clean interface. It is the primary integration surface for regulated custodians and other institutional partners.

9.4.7 AI model architecture

The Risk Scorer is a gradient-boosted decision tree ensemble (XGBoost, approximately 200 trees, depth 6) trained offline on historical FTSO price data, simulated vault states, and labelled risk events (drawdowns, liquidations). Features include: rolling volatility of FXRP/FLR/USD, vault leverage ratio, single-asset concentration, cross-chain exposure breakdown, recent hedge P&L, and time-since-last-rebalance. The model outputs a risk score (0–100) and a recommended action classification (hold, rebalance, hedge, deleverage). Training is done offline; the trained model is bundled into the TEE extension. The model is intentionally interpretable (SHAP values available) so that auditors can verify why a particular action was taken.

9.4.8 Inference pipeline

Inference runs entirely inside the TEE on each loop iteration. The PositionComputer rebuilds the current vault state from on-chain events and FDC attestations; the Risk Scorer runs on the feature vector derived from this state plus FTSO price history; the Policy Engine applies the deterministic policy constraints; the ActionExecutor emits any PMW instructions. The loop runs every FTSO round (approximately every 90 seconds) or on threshold events (large price moves, deposits, withdrawals). Latency budget: end-to-end loop under 5 seconds; PMW execution under 60 seconds; FDC attestation under 3 minutes.

9.4.9 Prompt architecture (for any LLM component)

If an LLM is used (for natural-language risk explanations in the auditor dashboard, not for risk decisions), it runs inside the TEE with a strict system prompt that forbids emitting position-level data. The prompt template is: 'You are an institutional risk analyst. Explain the following risk event in plain English for an auditor. Do not mention specific position sizes; refer only to aggregate metrics and policy thresholds. Event: {aggregate_event_summary}.' The LLM is a supplementary explainer, never a decision-maker.

9.4.10 Tool orchestration

The TEE orchestrates four tools: FTSO reader (price feeds), FDC verifier (external state attestation), PMW executor (cross-chain transactions), and SolvenyRoot publisher (on-chain proof publication). Each tool is invoked through a typed interface; the orchestrator enforces the loop order (observe → score → decide → act → attest). The orchestrator is deterministic: given the same inputs, it produces the same outputs, which is essential for verifiability.

9.4.11 Multi-agent coordination (if applicable)

The MVP uses a single agent per vault. A future multi-agent variant could run a RiskAgent, a HedgingAgent, and a ComplianceAgent in parallel inside separate TEE instances, with a Coordinator agent resolving conflicts. This is out of scope for the hackathon but is noted as a research direction.

9.5 Engineering Strategy

9.5.1 Frontend architecture

The institutional dashboard is a Next.js 16 application with App Router, TypeScript, Tailwind CSS, and shadcn/ui components. It connects directly to Flare RPC (for vault state) and to the Aegis FCC extension proxy (for attestation retrieval). Authentication is wallet-based (XRPL via Xaman, or EVM via MetaMask for auditors). The dashboard has three views: **Treasury** (depositor view: balances, recent actions, risk score), **Policy** (configurator for risk parameters), and **Audit** (auditor view: solvency proofs, verification tooling). The frontend is static-hostable (Vercel) with edge functions for attestation retrieval.

9.5.2 Backend architecture

There is no traditional backend. The Flare C-Chain contracts are the canonical state store; the FCC extension (Go, in TEE) is the computation layer; the dashboard is a thin client. This is a deliberate architectural choice: any stateful backend would undermine the verifiability claim. The only server-side component is the optional auditor-hosted attestation verification service, which is itself stateless and verifiable.

9.5.3 Infrastructure as Code

Smart contract deployment is managed via Foundry scripts (deploy.sh, with environment-specific configs for Coston2, Songbird, Mainnet). The FCC extension is containerised (Docker) and deployed via the Flare FCC deployment tooling. The dashboard is deployed via Vercel. All configuration is versioned in the GitHub repo. No manual configuration steps; the entire deployment is reproducible from a clean checkout.

9.5.4 CI/CD

GitHub Actions pipeline: on push, run Solidity tests (Foundry), Go tests (for the FCC extension), TypeScript tests (for the SDK and dashboard), and linting. On merge to main, deploy contracts to Coston2 (testnet), deploy the FCC extension to a testnet TEE, and deploy the dashboard to Vercel preview. Manual promotion to Songbird/Mainnet via tagged releases. The pipeline is itself open-source and visible in the repo, which contributes to the repository-quality score.

9.5.5 Testing strategy

- **Unit tests:** Solidity (Foundry, ~90% line coverage on vault contracts), Go (extension modules, ~80% coverage), TypeScript (SDK and dashboard utilities).
- **Integration tests:** end-to-end deposit → risk event → PMW execution → solvency attestation flow on Coston2, scripted and reproducible.
- **Fuzz tests:** Foundry fuzzing on vault contracts for edge cases (zero amounts, reentrancy, policy boundary conditions).
- **Demo dry-runs:** at least three full demo rehearsals on Coston2 before submission, with timing measured and recorded.
- **Failure-mode tests:** TEE unavailable, PMW consensus failure, FDC attestation delay — verify the vault enters safe state.

9.5.6 Monitoring, logging, observability

On-chain monitoring via the Flare block explorer and a custom Grafana dashboard querying Flare RPC. TEE-internal logging is minimal (the TEE should be deterministic), but structured logs of instruction receipt, policy decisions, and attestation publication are emitted to a read-only log stream. Auditor-facing observability is via the dashboard's Audit view. No metrics that could leak position information are published.

9.5.7 Reliability, fault tolerance, performance

Reliability target: 99.5% uptime during the demo window. Fault tolerance: the vault fails safe (no new positions) if the TEE is unavailable; users can always exit via `emergencyExit`. Performance: deposit confirmation under 30 seconds; risk loop under 5 seconds; PMW execution under 60 seconds; FDC attestation under 3 minutes; full audit verification under 5 minutes. These are achievable targets on Coston2 and acceptable for an institutional product.

9.5.8 Scalability

The architecture scales horizontally: each vault is an independent smart contract instance, and the FCC extension can serve multiple vaults. Bottlenecks are FDC attestation throughput (bounded by the round structure) and TEE compute capacity (one TEE instance can serve approximately 100 active vaults). For the hackathon, scale is irrelevant; for production, horizontal scaling is straightforward.

9.5.9 Cost optimisation

On-chain costs (gas) are minimised by batching PMW instructions and publishing solvency roots only on meaningful state changes (not every loop iteration). TEE compute costs are bounded by the loop frequency (every FTSO round, not continuously). Cloud costs are minimal (dashboard hosting only). The unit economics work at scale: at 50 bps management fee on USD 50M AUM, annual revenue is USD 250K, against infrastructure costs of low five figures.

9.5.10 Deployment strategy

MVP deployment target: Coston2 testnet (Flare's public testnet for Flare Mainnet). This is the verified-correct environment for FCC and PMW experimentation. Songbird deployment is a stretch goal if PMW is live there by the submission deadline. Flare Mainnet deployment is out of scope for the hackathon but is the first post-hackathon milestone.

9.5.11 Threat modelling

Primary threats: (1) **Smart contract exploit** — mitigated by OpenZeppelin libraries, Foundry fuzz tests, internal review, and timelocked multisig upgrades. (2) **TEE compromise** — mitigated by FCC attestation (TEE identity verified on-chain), deterministic extension logic, and the fact that the TEE cannot move funds outside policy constraints. (3) **AI agent misbehaviour** — mitigated by the deterministic Policy Engine that constrains agent actions to on-chain policy parameters. (4) **PMW key compromise** — mitigated by TEE key custody and data-provider consensus for signing. (5) **Frontend compromise** — mitigated by wallet-based authentication (no custodial keys held by the dashboard) and CSP/HSTS headers.

9.5.12 Security review and compliance

Internal security review during week 4 of the build (smart contracts) and week 5 (FCC extension). Post-hackathon: external audit (target Sigma Prime, OpenZeppelin, or Trail of Bits). Compliance: Aegis is positioned as infrastructure, not as a regulated entity; KYC/AML is delegated to integrating custodians (BitGo model). The FDC-based attestation of counterparty address validity supports the compliance workflow without making Aegis itself a regulated party.

9.6 Product and Business Strategy

9.6.1 Business model

Aegis charges a management fee (50 basis points annualised on AUM) and a performance fee (10% of yield above a benchmark). Fees accrue to the Aegis Treasury contract, which is governed by Aegis token holders (post-hackathon). The institutional SaaS tier adds a flat annual license for white-label deployment by regulated custodians, plus a per-vault setup fee.

9.6.2 Pricing strategy

Tier	Target user	Pricing	Notes
Retail (post-hackathon)	Individual XRP/FLR holders	50 bps mgmt + 10% perf	Self-serve via dashboard; no minimum.
Institutional	Treasuries USD 5M+	40 bps mgmt + 15% perf	Includes dedicated support, custom policies, auditor tooling.
White-label / SaaS	Regulated custodians	USD 100K/yr + per-vault fee	BitGo-style integration; custodian retains KYC/AML.
Infrastructure (open-source)	Other builders	Free (MIT license)	Builds ecosystem; monetises via paid support and hosted services.

9.6.3 Monetisation

Monetisation is layered: protocol fees (on-chain, automatic), institutional SaaS licenses (off-chain, contracted), and value-accruing token mechanics (post-launch, governed). The hackathon demo focuses on the protocol fee layer; the institutional and token layers are described in the roadmap but not implemented.

9.6.4 Go-to-market strategy

Phase 1 (hackathon → 3 months): secure two letters of interest from XRP-holding entities (target: a corporate treasury and a family office); publish the open-source vault contracts and SDK; build community via the Flare Hackathon Telegram group and Twitter. Phase 2 (3–9 months): launch on Flare Mainnet; onboard first institutional pilot; integrate with one regulated custodian (target: BitGo, given existing Flare partnership). Phase 3 (9–18 months): scale to 10+ institutional clients; launch white-label SaaS tier; raise a priced round from Flare-aligned investors.

9.6.5 Growth, adoption, community, ecosystem, partnership strategies

- **Growth:** content marketing around the verifiable-confidential treasury pattern; conference talks at Flare-dedicated and broader institutional crypto events; case studies from pilot institutions.
- **Adoption:** lower the institutional minimum over time as the protocol matures; open-source the vault contracts to enable community-audited deployments; subsidise the first auditor verification per vault.
- **Community:** Discord for developers; monthly office hours; bounty programme for FCC extension contributions.
- **Ecosystem:** integrate with Sceptre (sFLR), Clearstar (yield vaults), and Hyperliquid (hedging) as first-class strategy targets; list on the Flare Developer Hub as a reference FCC application.
- **Partnership:** prioritise BitGo (custody), VivoPower (pilot treasury), and EasyA (developer onboarding) as the first three named partners.

9.6.6 Open-source strategy

The vault contracts, the FCC extension, the TypeScript SDK, and the dashboard are open-source under MIT or Apache 2.0. The AI risk model weights are published under a permissive licence. The repository includes clear contribution guidelines, a code of conduct, and structured documentation. Open-sourcing is both an ecosystem play (building the Flare FCC developer community) and a competitive moat (community-audited code is harder for a closed-source competitor to match on trust).

9.6.7 Investor narrative

'Aegis is the institutional treasury layer for XRP-native DeFi on Flare. We use Flare's three newest primitives — FCC, PMW, and FDC — to deliver a product that has been impossible on any other chain: verifiable, confidential, AI-managed cross-chain treasury management. Our beachhead is the emerging corporate XRP treasury category, validated by VivoPower's USD 100M allocation. Our defensible moat is the compound integration of Flare primitives plus institutional trust and partner relationships. We are raising a priced round to scale from hackathon MVP to Mainnet launch with two paying institutional pilots.'

9.6.8 Sponsor narrative

'Aegis is the reference implementation of the Flare 2.0 thesis. It uses FCC for confidentiality, PMW for cross-chain execution, and FDC for verifiable solvency — the three pillars of the Flare 2.0 vision — in a single institutional product. It targets the customer segment Flare's commercial team is actively courting (VivoPower-class treasuries). It generates on-chain activity that flows through the FLR burn flywheel. It is the project Flare can point to when asked: what is FCC actually for?'

9.6.9 Judge narrative

'Aegis is the only submission that uses all five Flare primitives in a load-bearing way. Remove Flare and the product cannot exist. The demo shows a private vault, an AI agent rebalancing positions inside a TEE, cross-chain execution via PMW, and a verifiable solvency proof — all in under five minutes. The team has a clear institutional customer, a credible business model, and a path to Mainnet. This is the project that tells the Flare 2.0 story.'

9.6.10 Defensibility analysis

Defensibility is compound. **Technical:** the FCC + PMW + FDC integration is non-trivial to replicate; the open-source vault contracts become the reference implementation others build on. **Commercial:** institutional trust accumulates slowly; the first auditor verification workflow becomes the standard. **Network:** institutions prefer the treasury protocol that other institutions use (shared auditors, shared compliance). **Brand:** being the 'first FCC institutional app' is a category-defining position that is hard to displace.

9.7 Execution Roadmap

9.7.1 MVP scope

The MVP delivers: (1) a VaultCore contract on Coston2 that accepts FXRP deposits and enforces a simple policy; (2) an FCC extension with PositionComputer and SolvencyAttestor modules; (3) a basic AI risk agent that reads FTSO and emits a rebalance instruction on a threshold breach; (4) a PMW-mediated cross-chain flow (XRPL settlement) for the rebalance; (5) an auditor verification flow via the SolvencyRoot contract; (6) an institutional dashboard with Treasury, Policy, and Audit views; (7) an open-source repo with README, architecture docs, and deployment scripts.

9.7.2 Development phases

Phase	Dates	Focus	Deliverable
P0 — Foundation	Jun 29 – Jul 5 (week 1)	Repo setup, FCC scaffold running on Coston2	Working PMW flow; VaultCore deployment on Coston2; PMW test transaction confirmed end-to-end
P1 — Vault contracts	Jul 6 – Jul 12 (week 2)	VaultCore, PolicyRegistry, SolvencyRoot, Indexer	Deployed contracts; Verified Role2; Foundry tests passing; first full demo rehearsal completed
P2 — FCC extension	Jul 13 – Jul 19 (week 3)	PositionComputer, RiskAgent (XGBoost), SolvencyAttestor, Auditor	AI risk model; AI agent; AI policy; AI position; AI rebalance + attestation flows
P3 — PMW + FDC integration	Jul 20 – Jul 26 (week 4)	PMW XRPL flow; FDC attestation integration	Full end-to-end flow: PMW deposit → PMW rebalance → solvency root → FDC attestation
P4 — Frontend	Jul 27 – Aug 2 (week 5)	Institutional dashboard (Treasury, Policy, Audit views); SDK	Dashboard live; SDK published; demo flow rehearsed end-to-end
P5 — Polish & demo	Aug 3 – Aug 10 (week 6)	Documentation, video, pilot outreach, demo rehearsals, contingency plan	Submission package complete; 3+ demo rehearsals; LOI from at least one institutional prospect
P6 — Submit	Aug 11 – Aug 14	Final submission, judge-ready demo, FAQ responses	Submission dropped; Demo Hacks; demo video live; repo public

9.7.3 Milestones

- **M1 (end of week 1):** FCC scaffold extension running on Coston2; PMW test transaction confirmed end-to-end. Go/no-go decision on PMW scope.
- **M2 (end of week 2):** Vault contracts deployed and tested on Coston2; policy enforcement verified.
- **M3 (end of week 3):** FCC extension processing deposit + rebalance + attestation flows with mock PMW.
- **M4 (end of week 4):** Full end-to-end flow with real PMW and FDC attestation on Coston2. Demo path proven.
- **M5 (end of week 5):** Dashboard live; SDK published; first full demo rehearsal completed.
- **M6 (end of week 6):** Submission package complete; LOI from at least one institutional prospect; demo rehearsed three times; contingency plan tested.

9.7.4 Risk mitigation

- **PMW risk:** if PMW is not workable by end of week 1, fall back to a simulated cross-chain flow with explicit labelling in the demo and documentation. The verifiable solvency story still stands.
- **FCC complexity risk:** if the FCC extension is harder than expected, scope the AI risk model down to a deterministic rule engine (no XGBoost) and label the ML component as a stretch goal.
- **Pilot risk:** if no LOI is secured, proceed without one; the demo works with test funds and the institutional narrative is still credible.
- **Demo failure risk:** pre-recorded fallback video; live demo on Coston2 with cached state to minimise network dependency.
- **Team capacity risk:** cut the SDK scope first, then the multi-vault support, then the auditor dashboard — in that order — if behind.

9.7.5 Stretch goals

- Songbird deployment (in addition to Coston2) if PMW is live there.
- Multi-vault support (single contract serving multiple institutions).
- Natural-language risk explanation via an in-TEE LLM (supplementary only).
- Real auditor pilot: a Big Four firm's forensics team verifies a vault.

- Integration with BitGo's API for the institutional compliance surface.

9.7.6 Post-hackathon roadmap

Month 1: external security audit; Mainnet deployment preparation. Month 2–3: first institutional pilot onboarding; Mainnet launch. Month 4–6: second and third institutional pilots; BitGo integration; open-source community launch. Month 7–12: scale to 10+ institutions; raise priced round; expand to additional asset types (e.g., FDOGE, FBTC when supported).

9.7.7 Scaling strategy

Scaling is horizontal: more vaults, more institutions, more asset types, more chains (as PMW adds support). The architecture supports this without redesign. The principal scaling constraint is institutional trust acquisition, which is a sales and partnership function rather than a technical one.

9.7.8 Grand Prize advantage

Aegis dominates the judging rubric on the dominant criterion (Flare integration quality) by being the only submission that uses all five Flare primitives in a load-bearing way. It outperforms the expected competitor categories (Section 7) on every dimension that matters: AI onboarding clones lack primitive depth; FXRP wrappers lack FCC and PMW; FCC Hello World extensions lack product framing; simple private treasury entries lack PMW, FDC, and institutional narrative. The wow moments are: (1) the deposit flow that mints FXRP from a single XRPL signature; (2) the AI agent autonomously rebalancing via PMW across chains; (3) the auditor verifying solvency without ever seeing positions. The assumptions that must be validated during development are: (a) FCC on Coston2 is stable enough for a non-trivial extension; (b) PMW supports the XRPL flows needed; (c) FDC attestation latency is acceptable; (d) the judging panel includes technical depth.

10. Engineering Execution Strategy — Day-by-Day

This section translates the roadmap of Section 9.7 into a day-level execution plan, optimised for maximum judging impact, fastest delivery, lowest unnecessary complexity, and demo reliability.

Day	Date	Primary task	Output / acceptance criterion
1	Jun 29	Repo bootstrap; clone FCC scaffold; deploy Hello-World extension to Coston2.	Scaffold on Coston2; extension attests on Coston2; README skeleton up.
2	Jun 30	Validate PMW on Coston2: test XRPL wallet creation, instruction signing.	On-chain PMW-mediated XRPL transaction confirmed from a Flare contract.
3	Jul 1	FDC integration spike: request XRPPayment attestation; verify proof.	FDC attestation retrieved and verified from the extension.
4	Jul 2	Design vault contracts (VaultCore, PolicyRegistry, SolvencyRoot, InstructionSender).	Substitution finalised; reviewed by team.
5	Jul 3	Implement VaultCore + PolicyRegistry; Foundry tests.	Deposit + policy enforcement tested on local anvil.
6	Jul 4	Implement SolvencyRoot + InstructionSender + VerifierRole.	All five contracts deployed on Coston2; fuzz tests green.
7	Jul 5	M1 checkpoint; end-to-end walk-through of contracts on Coston2.	M1 sign-off; PMW go/no-go decision.
8	Jul 6	Build PositionComputer module (rebuild state from on-chain events).	PositionComputer returns correct vault state on test events.
9	Jul 7	Build SolvencyAttestor module (Merkle root computation + publication).	SolvencyRoot published on-chain from extension.
10	Jul 8	Train XGBoost risk model on historical FTSO data (offline).	Model file bundled into extension; inference runs in TEE.
11	Jul 9	Build RiskAgent module (loop: observe → score → decide → act).	Agent tests full loop on Coston2 with mock PMW.
12	Jul 10	Build ActionExecutor + Policy Engine (deterministic policy enforcement).	Policy constraints enforced; agent cannot exceed limits.
13	Jul 11	M2 checkpoint; full FCC extension processing deposit + rebalance.	M2 sign-off.
14	Jul 12	PMW integration: wire ActionExecutor to PMW for XRPL execution.	Agent triggers real PMW XRPL transaction on policy breach.
15	Jul 13	FDC integration: attestation of XRPL payment + Hyperliquid state.	External state attested and fed back to PositionComputer.
16	Jul 14	End-to-end flow: deposit → risk event → PMW rebalance → solvency flow.	Vault flow tested on Coston2; recorded as demo seed.
17	Jul 15	Error handling, safe-state logic, emergency exit.	Failure-mode tests pass (TEE down, PMW failure, FDC delay).
18	Jul 16	M3 checkpoint; demo path proven end-to-end.	M3 sign-off; demo script v1 drafted.
19	Jul 17	Frontend: dashboard scaffold (Next.js, shadcn/ui, wallet auth).	Dashboard connects to Flare RPC and FCC extension proxy.
20	Jul 18	Frontend: Treasury view (balances, recent actions, risk score).	Depositor can see vault state and recent rebalances.
21	Jul 19	Frontend: Policy view (configurator).	Depositor can set and inspect risk policies.
22	Jul 20	Frontend: Audit view (solvency proofs, verification tooling).	Auditor can request and verify a solvency attestation.
23	Jul 21	TypeScript SDK (VaultClient, PolicyClient, AuditClient).	SDK published; example script demonstrates full flow.
24	Jul 22	M4 checkpoint; first full demo rehearsal.	M4 sign-off; demo timing under 5 minutes.
25	Jul 23	Documentation: README, architecture doc, deployment guide.	Docs complete; repo is self-explanatory.
26	Jul 24	Demo video v1 recorded; submission draft on DoraHacks.	Video + submission draft ready for review.
27	Jul 25	Pilot outreach: send Aegis summary to 5 institutional prospects.	2 conversations initiated.
28	Jul 26	Demo rehearsal #2; refine script based on timing and clarity.	Demo script v2; timing optimised.
29	Jul 27	Hardening: more Foundry fuzz tests; edge case coverage.	Test coverage > 85% on vault contracts.
30	Jul 28	Polish: UI/UX pass; loading states; error messages; copy.	Dashboard looks production-grade.
31	Jul 29	M5 checkpoint; third demo rehearsal; LOI secured (target).	M5 sign-off; LOI in hand (or fallback narrative ready).
32	Jul 30	Contingency plan: pre-recorded demo + offline fallback.	Backup demo tested.
33	Jul 31	Judge Q&A prep: draft answers to 20 likely technical questions.	Q&A doc complete.
34	Aug 1	Final repo cleanup; CI green; all tests pass.	Repo is judge-ready.
35	Aug 2	Final demo rehearsal #4; record final demo video.	Final demo video published.
36–40	Aug 3–7	Buffer: address issues found in rehearsals; final polish; partner conversations.	No critical issues open.
41	Aug 8	Pre-submission review: every rubric criterion explicitly addressed.	Submission checklist complete.

42	Aug 9	Submit on DoraHacks; verify all links and contracts.	Submission live and verified.
43–46	Aug 10–13	Judge Q&A window: respond promptly to any judge questions.	All judge questions answered within 24h.
47	Aug 14	Final submission deadline.	Done.

11. Winning Demo and Presentation Strategy

11.1 Demo script — five-minute live demo

0:00–0:30 — Thesis. 'Institutional XRP treasuries are emerging — VivoPower committed USD 100M in June 2025. They need three things no Flare product offers today: confidentiality, verifiable solvency, and autonomous cross-chain risk management. Aegis delivers all three using Flare's newest primitives — FCC, PMW, and FDC — together.'

0:30–1:15 — Deposit (Layer 1). Switch to the dashboard. Sign in with the XRPL wallet (Xaman). Sign a single XRPL transaction that mints FXRP via Flare Smart Accounts and deposits into the Aegis vault. Show the FDC attestation confirming the XRPL payment. Show the vault balance updating on-chain. 'One signature, one on-chain deposit, fully attested.'

1:15–2:30 — Confidential Position (Layer 3). Open the vault view. Show that the on-chain state shows only the deposit and a Merkle root — not the full position. 'The full position — what we hold, where, and what we owe — is computed inside this TEE.' Show the TEE attestation proof. 'Anyone can verify the TEE ran the correct code; no one can see the positions inside it.'

2:30–3:30 — Autonomous Risk Rebalance (Layers 3 + 4). Trigger a simulated market drawdown (or wait for one if timing permits). Show the AI risk agent — running inside the TEE — detect the threshold breach, compute a rebalance action, and issue a PMW instruction. Show the PMW signing flow (data-provider consensus) and the resulting XRPL transaction. Show the FDC attestation of the executed payment. Show the updated solvency root published on-chain. 'An AI agent inside a TEE just autonomously rebalanced a private vault across chains, and every step is verifiable.'

3:30–4:30 — Verifiable Solvency (Layer 5). Switch to the auditor view. Request a fresh solvency attestation. Show the TEE computing a proof that assets exceed liabilities by a stated margin, without revealing the underlying amounts. Verify the proof on-chain. 'An auditor can verify this treasury is solvent without ever seeing a single position. That is the confidentiality-to-verifiability transformation — and it is only possible on Flare.'

4:30–5:00 — Close. 'Aegis uses every Flare primitive in a load-bearing way. We have an institutional pilot conversation underway [or: LOI from X]. We will launch on Mainnet within ninety days of the hackathon. The repo, SDK, and documentation are public. This is the reference implementation of the Flare 2.0 thesis — and we would value your questions.'

11.2 Demo contingency plan

- **Live demo fails:** switch to pre-recorded video of the same flow, narrated live. State explicitly: 'For reliability, here is the same flow recorded on Coston2 yesterday; the live version is also available if you would like to see it after the Q&A.'
- **PMW transaction fails:** skip the live PMW step, show the pre-recorded PMW execution, and emphasise the FDC-attested result on the dashboard.
- **FDC attestation slow:** pre-fetch an attestation before the demo and use the cached version; explain that the production flow is identical.
- **TEE attestation fails to verify:** this is unlikely (the TEE identity is registered on-chain), but if it happens, show the attestation log and offer to walk through the verification step by step in the Q&A;
- **Dashboard does not load:** have the dashboard running locally on a second machine as a hot standby.
- **Network issue:** have the demo video on a USB stick and a mobile hotspot.

11.3 Pitch decks

11.3.1 Investor-grade pitch deck (10 slides)

- Slide 1: Title — Aegis, the institutional treasury layer for Flare.

- Slide 2: Problem — the four institutional gaps (confidentiality, verifiable solvency, cross-chain execution, autonomous risk).
- Slide 3: Solution — Aegis in one sentence + the five-layer architecture.
- Slide 4: Why Flare — the five primitives, each load-bearing.
- Slide 5: Market — XRP treasury category (VivoPower + pipeline).
- Slide 6: Product — dashboard screenshots + demo flow.
- Slide 7: Business model — fees + SaaS + token.
- Slide 8: Traction — LOI, pilot conversations, repo stats.
- Slide 9: Roadmap — hackathon to Mainnet to scale.
- Slide 10: Team + ask.

11.3.2 Technical presentation (for technical judges)

- Architecture deep-dive: the five layers with each Flare primitive's role.
- FCC extension walkthrough: PositionComputer, RiskAgent, SolvencyAttestor.
- PMW flow: data-provider consensus, TEE key custody, XRPL execution.
- FDC attestation: which attestation types, what latency, what verification.
- Security model: smart contract, TEE, economic, and AI safety layers.
- Verifiability invariant: how an auditor reconstructs state from on-chain data.
- Repo tour: contracts, extension, SDK, tests, deployment scripts.

11.3.3 Sponsor presentation (for Flare team)

- Why this advances Flare 2.0: FCC + PMW + FDC reference implementation.
- Why this serves Flare's commercial priorities: institutional (BitGo, VivoPower), verifiable AI, cross-chain execution.
- Why this drives FLR value accrual: FAssets minting volume, Smart Account usage, gas burns.
- What we need from Flare: intro to BitGo; visibility on the Flare Developer Hub; consideration for a follow-on grant.
- Open-source contribution: vault contracts, SDK, and FCC extension patterns available to the Flare community.

11.4 One-minute pitch

'Aegis is the institutional treasury layer for XRP-native DeFi on Flare. We use Flare's three newest primitives — FCC for confidentiality, PMW for cross-chain execution, and FDC for verifiable solvency — to deliver a product no other chain can: a private, AI-managed treasury that any auditor can verify without seeing positions. Our beachhead is the emerging corporate XRP treasury category — VivoPower committed USD 100M in June. We are the reference implementation of the Flare 2.0 thesis. Repo and SDK are public; Mainnet launch in ninety days.'

11.5 Three-minute pitch

Extend the one-minute pitch with: the four institutional pain points in one sentence each; the five-layer architecture in one sentence each; the demo wow-moments (deposit, autonomous rebalance, auditor verification) described but not yet shown; the business model in one sentence; the traction in one sentence; the ask in one sentence.

11.6 Judge Q&A; preparation — likely questions and answers

Question	Suggested answer (concise)
Why not build this on Ethereum or Base?	None of them enshrine FCC, PMW, and FDC together. The product depends on all three; remove Flare and
How do you handle TEE compromise?	TEE identity is registered and verified on-chain; the extension logic is deterministic; PMW requires data-p
What if the AI agent makes a bad decision?	The Policy Engine is deterministic and constrained by on-chain parameters; the agent cannot exceed policy
How do you acquire institutional customers?	Named partner outreach (BitGo, VivoPower); Flare introductions; conference presence; pilot programme v
Is this a regulated activity?	Aegis is infrastructure; regulated custodians integrate for the regulated surface; KYC/AML is delegated vi
What is your business model?	50 bps management fee + 10% performance fee on protocol; USD 100K/yr SaaS license for white-label cu
How do you verify solvency without seeing positions?	Flare computes the solvency margin and publishes a Merkle root; the auditor verifies the proof cryptograph
Why is FDC necessary?	FDC attests external chain state (XRPL payments, Base OFT transfers, address validity) so the TEE can re
How is this different from BridgeSafe or other private mainframes?	We use PMW for cross-chain execution and FDC for verifiable solvency, which simpler entries do not; we
What is the hardest part you built?	The FCC extension that combines position computation, AI risk inference, and solvency attestation in one
What would you do with a Flare grant?	External audit, Mainnet deployment, first institutional pilot, BitGo integration.
How do you prevent the AI from being manipulated?	Flare is a FTSO and economically secured by Flare consensus; the Risk Agent uses FTSO as one input

11.7 Objection handling strategy

- **'This is too ambitious for a hackathon.'** 'Every component is demonstrated working on Coston2 today. We scoped aggressively: one vault, one policy, one rebalance flow, one auditor verification. The architecture scales; the demo is focused.'
- **'The AI is just a wrapper around an LLM.'** 'The Risk Scorer is an XGBoost model running inside the TEE, not an LLM call. The Policy Engine is deterministic. The LLM, where used, is for natural-language explanations to auditors, not for decisions.'
- **'PMW is too new to be reliable.'** 'We validated the PMW flow on Coston2 in week one. We have a fallback path (simulated PMW with explicit labelling) if the live flow fails during the demo. The verifiable solvency story does not depend on PMW.'
- **'There is no real institutional customer yet.'** 'We have [LOI / active conversation] with [X]. The product works with test funds today; the institutional pilot is a commercial milestone, not a technical dependency.'
- **'Why not just use Aztec / a zk chain?'** 'zk chains provide privacy but not verifiable cross-chain execution or FDC-style external state attestation. The composition of confidentiality, verifiability, and cross-chain execution is unique to Flare.'

12. Red-Team Critique and Refinement

This section assumes the role of a competing Grand Prize team attempting to defeat Aegis. We identify the project's most significant weaknesses across technical, product, judging, and execution dimensions, and then redesign the project to eliminate or reduce each weakness. The refinement loop is repeated until further improvements become marginal.

12.1 Round 1 — Initial critique

12.1.1 Technical weaknesses

- **W1 (High):** PMW is brand new and may have undocumented constraints or rough edges that surface only during integration. Mitigation: validate PMW on Coston2 in week one; design a simulated-PMW fallback path; explicitly label the live PMW component in the demo so judges can distinguish it from the simulated flow.
- **W2 (High):** FCC on Coston2 may not be production-stable; extension crashes during the demo would be catastrophic. Mitigation: extensive dry-runs; pre-recorded fallback; safe-state logic that preserves vault funds even if the TEE fails.
- **W3 (Medium):** FDC attestation latency (typically minutes per round) may make the live demo feel slow. Mitigation: pre-fetch attestations before the demo; use cached attestations with explicit labelling; rehearse timing.
- **W4 (Medium):** The XGBoost risk model is a black box; judges may question its trustworthiness. Mitigation: publish SHAP values in the auditor dashboard; emphasise that the Policy Engine is deterministic and constrains the model; offer to walk through a specific decision in the Q&A.
- **W5 (Low):** Smart contract risk in the vault contracts. Mitigation: OpenZeppelin libraries, Foundry fuzz tests, internal review, timelocked multisig.

12.1.2 Product weaknesses

- **W6 (High):** No real institutional customer at submission time. Mitigation: aggressive pilot outreach in week 5; target at least one LOI; if no LOI, emphasise the VivoPower partnership as market validation and present the product as ready for the first pilot.
- **W7 (Medium):** The institutional dashboard may look like a demo rather than a product. Mitigation: invest in UI/UX polish in week 5; use real-looking data; include loading states and error messages that suggest production maturity.
- **W8 (Medium):** The product is not deployable on Mainnet within the hackathon (FCC and PMW are on Coston2). Mitigation: be explicit about the Coston2 deployment; present the Mainnet path as a 90-day post-hackathon milestone; do not overclaim.
- **W9 (Low):** Single-vault MVP may seem narrow. Mitigation: frame the vault as a generalisable pattern; include a one-paragraph extension section in the README showing how the pattern applies to lending, fund administration, etc.

12.1.3 Judging weaknesses

- **W10 (Medium):** The judging panel may not include a technical judge capable of appreciating the FCC + PMW + FDC composition. Mitigation: write the README and demo script so that the integration depth is legible to a non-technical judge without dumbing it down for a technical one. Use the 'remove Flare and the product cannot exist' framing prominently.
- **W11 (Medium):** The demo has many moving parts; if any one fails, the impression suffers. Mitigation: rehearse the demo at least four times; have a pre-recorded fallback; structure the demo so that each section stands alone (deposit can be shown without PMW; PMW can be shown without audit; etc.).
- **W12 (Low):** Some judges may prefer a more consumer-friendly concept. Mitigation: include a brief 'retail future' slide showing how the vault pattern extends to individual XRP holders; do not let this dilute the institutional focus.

12.1.4 Execution risks

- **W13 (High):** Six weeks is tight for the full scope. Mitigation: aggressive MVP scoping; daily standups; cut features in the priority order: SDK first, then multi-vault, then auditor dashboard, then ML component.
- **W14 (Medium):** Team capacity (assumed 2–3 people). Mitigation: clear role ownership; pair-program the FCC extension and the smart contracts in parallel; use the day-by-day plan.
- **W15 (Low):** Burnout risk in the final week. Mitigation: buffer days (36–40) in the schedule; no all-nighters in week 6; submit at least 24 hours before the deadline.

12.1.5 Hidden assumptions

- **A1:** FCC on Coston2 supports the extension pattern in the quickstart without significant deviation. Validate in week 1.
- **A2:** PMW on Coston2 supports XRPL wallet creation and instruction signing. Validate in week 1.
- **A3:** FDC supports XRPPayment attestation with acceptable latency. Validate in week 1.
- **A4:** The judging panel includes at least one technical judge. Mitigation: assume not, and write for a mixed audience.
- **A5:** An institutional LOI is achievable in five weeks. Mitigation: have a fallback narrative if not.

12.1.6 Scalability, security, and business risks

- **S1 (Scalability):** FDC attestation throughput may limit multi-vault scaling. Mitigation: batch attestations; publish solvency roots on meaningful state changes only.
- **S2 (Security):** A subtle bug in the Policy Engine could let the AI agent exceed policy limits. Mitigation: deterministic Policy Engine with on-chain parameter enforcement; fuzz test the policy boundary conditions extensively.
- **S3 (Business):** Institutional sales cycles are long; revenue may lag. Mitigation: layer revenue (protocol fees from retail first, institutional SaaS later); pursue grant funding to bridge.

12.2 Round 2 — Refined design

Applying the mitigations produces a refined design with three structural changes from the initial concept:

- **Change 1:** Add an explicit simulated-PMW fallback path with clear on-screen labelling, so the demo never fails entirely if the live PMW flow has issues. The fallback is presented honestly as a fallback, not as the primary path.
- **Change 2:** Add SHAP-based model explainability to the auditor dashboard, so any AI decision can be traced to its contributing features. This converts a perceived weakness (black-box AI) into a strength (auditable AI).
- **Change 3:** Add a 'retail future' slide and a one-paragraph extension section in the README, addressing the W12 concern without diluting the institutional focus.

12.3 Round 3 — Marginal improvements

After Round 2, the remaining improvements are marginal: additional fuzz testing, more polished UI micro-interactions, a more detailed post-hackathon roadmap. None of these change the fundamental design; they are execution refinements. The refinement loop terminates here.

12.4 Residual risks after refinement

- PMW may simply not work end-to-end on Coston2 within the build window. This is the single largest residual risk; it is mitigated but not eliminated by the fallback path.
- No institutional LOI may materialise. The narrative holds without one, but the 'clarity and future potential' score is stronger with one.

- A competing team may submit a similar concept with deeper PMW integration. The defence is the compound integration (FCC + FDC + FAssets + PMW + FTSO) plus the institutional framing, which is difficult to replicate in six weeks.

13. GitHub and Documentation Strategy

13.1 Repository structure

```
aegis/ README.md # thesis, architecture, demo, deployment LICENSE # MIT CONTRIBUTING.md
CODE_OF_CONDUCT.md docs/ architecture.md # five-layer architecture, diagrams
deployment.md # Coston2 / Songbird / Mainnet deployment security.md # threat model,
mitigations, audit status api.md # contract + extension API reference demo-script.md # the
five-minute demo script contracts/ foundry.toml src/ VaultCore.sol PolicyRegistry.sol
SolvencyRoot.sol InstructionSender.sol VerifierRole.sol interfaces/... test/
VaultCore.t.sol PolicyRegistry.t.sol ... # fuzz tests, integration tests script/
Deploy.s.sol DeployCoston2.s.sol extension/ # FCC extension (Go, in TEE) go.mod
cmd/server/main.go internal/ position/ # PositionComputer risk/ # RiskAgent (XGBoost)
policy/ # Policy Engine attestation/ # SolvencyAttestor executor/ # ActionExecutor (PMW)
Dockerfile deployment/ # FCC deployment configs sdk/ # TypeScript SDK src/ vault-client.ts
policy-client.ts audit-client.ts test/... package.json frontend/ # Next.js dashboard app/
treasury/ policy/ audit/ components/... examples/ deposit-flow.ts audit-verify.ts
.github/workflows/ ci.yml # Foundry + Go + TypeScript tests deploy-coston2.yml
```

13.2 README structure

The README is the single most-read artifact. It must communicate the thesis, the Flare integration depth, the demo path, and the deployment status — all in the first screen. Recommended structure: (1) Project title and one-sentence thesis; (2) Why this wins (three bullets: uses all five Flare primitives; institutional ICP; verifiable-confidential pattern); (3) Demo video link + live dashboard link; (4) Architecture diagram; (5) How Flare is used (a table mapping each primitive to its load-bearing role); (6) Quickstart (deploy on Coston2 in under 10 minutes); (7) Repository structure; (8) Roadmap; (9) License and contributing.

13.3 The 'How Flare is used' table

Flare primitive	Load-bearing role in Aegis	What breaks if removed
FAssets (FXRP)	Core asset deposited into vaults.	No asset to manage.
FTSO	Price feeds for risk scoring and policy thresholds.	Risk agent cannot detect drawdowns.
FDC	Attests external chain state (XRPL payments, address validity) for composition and deposit.	Cannot compose or deposit chain state verifiably.
FCC	TEE-based position computation and AI risk inference; publishes solvency proofs.	No verifiable solvency.
PMW	Cross-chain execution (XRPL settlement, Base OFT, Hyperliquid hedging).	No cross-chain rebalancing.
Smart Accounts	Onboards XRP holders from XRPL wallets with a single step.	Onboarding friction returns.

This table is the single most powerful artifact in the README. It directly answers the judging criterion 'Flare integration quality' and makes the 'remove Flare and the product cannot exist' argument visible at a glance.

13.4 Documentation depth

- **Architecture doc:** the five layers, each primitive's role, data flow, sequence diagrams (deposit, rebalance, audit).
- **Deployment doc:** step-by-step Coston2 deployment; Songbird and Mainnet roadmap; environment variables and configuration.
- **Security doc:** threat model, mitigations, audit status, responsible disclosure policy.
- **API doc:** contract interfaces, extension HTTP API, SDK reference, examples.
- **Demo script doc:** the five-minute demo script with timings, contingency plans, and Q&A; prep.

13.5 Open-source practices

- MIT or Apache 2.0 licence; permissive to maximise adoption.

- Clear CONTRIBUTING.md with code review requirements.
- Code of Conduct (Contributor Covenant).
- Semantic versioning for releases.
- GitHub Actions CI on every push and PR.
- Issue templates (bug, feature, security).
- Pull request template (summary, testing, breaking changes).
- Discussions enabled for community Q&A.;

14. Security and Risk Assessment

14.1 Threat model

Threat	Vector	Impact	Mitigation	Residual risk
Smart contract exploit	Bug in vault or policy contract.	Loss of deposited funds.	OpenZeppelin libs, Foundry fuzz, internal review, timelocked multisig.	Medium (pre-launch).
TEE compromise	TEE vulnerability or side-channel attack.	Position disclosure or false attestation.	Fail-safe attestation, deterministic logic, frequent updates.	Low (if TEE model is mature).
AI agent misbehaviour	Model error or adversarial input.	Excessive rebalancing; loss of funds.	Deterministic Policy Engine, on-chain policy constraints, fail-safe.	Low.
PMW key compromise	TEE key extraction.	Unauthorised cross-chain transfers.	TEE key custody, data-provider consensus, cosigners.	Low.
FDC attestation delay	Network congestion or provider unavailability.	State failure.	Safe-state logic; cached attestations; fallback to last-known-good.	Low.
Frontend compromise	XSS, supply chain, DNS hijacking.	Phishing; false instructions.	CSP/HSTS, SRI, wallet-based auth (no custodial keys), signed releases.	Medium.
Economic attack	Manipulation of FTSO or value-at-risk signals.	Uncontrolled risk signals.	FTSO is enshrined and economically secured.	Low.
Governance attack	Malicious multisig signer.	Unauthorised upgrade.	Timelocked upgrades; multisig threshold; public governance forum.	Medium.
Regulatory action	Classification of Aegis as regulated activity.	Operational disruption.	Position as infrastructure; partner with regulated custodians; legal counsel.	Medium.

14.2 Compliance considerations

Aegis is positioned as decentralised infrastructure, not as a regulated financial entity. KYC/AML obligations attach to the integrating custodian (e.g., BitGo), not to Aegis itself. FDC's AddressValidity attestation type supports the compliance workflow by allowing the vault to refuse deposits from non-verified addresses. The auditor verification flow supports periodic regulatory reporting without compromising confidentiality. Legal counsel should review the specific jurisdictional classification before Mainnet launch.

14.3 Risk register — top five

- 1. PMW integration risk** (High likelihood, High impact): PMW is new; integration may surface undocumented constraints. Owned by: lead engineer. Mitigation: week-1 validation; simulated fallback.
- 2. Pilot acquisition risk** (Medium likelihood, Medium impact): no LOI by submission. Owned by: founder. Mitigation: outreach in week 5; fallback narrative.
- 3. Scope creep risk** (Medium likelihood, Medium impact): team attempts too much. Owned by: PM. Mitigation: aggressive MVP scope; cut priority list (SDK, multi-vault, auditor dashboard, ML).
- 4. Demo failure risk** (Low likelihood, High impact): live demo fails in front of judges. Owned by: demo lead. Mitigation: pre-recorded fallback; four rehearsals; cached attestations.
- 5. Smart contract risk** (Low likelihood, High impact): vulnerability in vault contracts. Owned by: lead engineer. Mitigation: OpenZeppelin, Foundry fuzz, internal review, timelocked multisig; external audit post-hackathon.

15. Final Decision Matrix

15.1 Weighted scoring matrix — Top 3 concepts

Criterion (weight)	Aegis	Concept 2 (AI credit)	Concept 3 (PMW router)
Flare integration depth (15%)	5	4	4
Product usefulness (12%)	5	4	3
Ecosystem alignment (10%)	5	4	5
Sponsor alignment (8%)	5	3	4
Technical feasibility, 6 weeks (10%)	3	4	3
Differentiation (8%)	5	4	5
Business potential (7%)	5	4	3
User value (6%)	5	4	3
Scalability (5%)	4	4	4
Demo potential (8%)	5	3	4
Documentation potential (3%)	5	4	4
Open-source potential (3%)	5	4	5
Investment potential (3%)	5	4	3
Judge excitement (2%)	5	4	4
Weighted total (out of 100)	93	88	84

Scores: 5 = excellent, 4 = strong, 3 = acceptable, 2 = weak, 1 = poor. Weighted total = sum of (score × weight) × 20, normalised to 100.

15.2 Confidence levels

Recommendation	Confidence	Key dependencies
Aegis is the highest-probability Grand Prize concept	High (75–85%)	FCC and PMW work on Coston2; team executes the day-by-day plan; de
Aegis wins Bounty 2 (Confidential Compute Apps) outright	High (70–80%)	Same as above; no competing entry uses PMW + FDC + FCC + FAssets
Aegis places top 3 in Bounty 1 (Interoperable Asset Platform)	Medium-High (55–70%)	FAssets integration is strong but not the project's primary axis.
Aegis captures the Grand Prize	Medium-High (55–70%)	Assumes the Grand Prize exists and rewards cross-bounty excellence; de
Aegis leads to a post-hackathon institutional pilot	Medium (45–60%)	Depends on outreach success; VivoPower and BitGo are the primary targ
Aegis becomes a fundable startup	Medium (50–60%)	Depends on team, market timing, and post-hackathon execution.

15.3 Key assumptions

- FCC on Coston2 supports a non-trivial extension of the kind Aegis requires (Validated: verified against the Developer Hub quickstart).
- PMW on Coston2 supports XRPL wallet creation and instruction signing (To validate: week 1 spike).
- FDC supports the required attestation types with acceptable latency (Verified: documentation confirms the attestation types; latency to be measured).
- The judging panel applies the published rubric in good faith and rewards meaningful Flare integration (Reasoned assumption: based on prior winner patterns).
- A team of 2–3 competent engineers can deliver the MVP in six weeks (Reasoned assumption: based on the day-by-day plan and aggressive scoping).

- At least one institutional prospect will engage in conversation (Reasoned assumption: VivoPower's public commitment suggests category interest).

15.4 Trade-offs accepted

- **Depth over breadth:** the project is deep in the institutional treasury vertical rather than broad across multiple verticals. This maximises Flare integration depth at the cost of total addressable market in the short term.
- **Coston2 over Mainnet:** deploying on Mainnet within the hackathon is infeasible (FCC and PMW are testnet-only). The trade-off is a credible Coston2 deployment with an explicit Mainnet path, rather than an overclaimed Mainnet deployment.
- **Institutional focus over retail appeal:** the demo targets judges and institutional prospects rather than the broad retail audience. This may slightly reduce 'judge excitement' from non-technical judges but maximises the dominant criterion (Flare integration quality).
- **Real PMW with fallback over simulated PMW only:** the team invests in real PMW integration with an explicit fallback path, rather than presenting a simulated flow as the primary path. This is honest and higher-ceiling, but higher-risk.

15.5 Alternative concepts if constraints change

Constraint change	Recommended alternative	Why
PMW proves infeasible on Coston2 by week 12	Concept 12: Verifiable AI credit/underwriting engine	Engines PMW + FCC + FAssets; still strong on Flare
FCC proves too unstable for non-trivial concepts	Concept 10: FDC-based proof-of-reserves service (BIP6)	Per (BIP6) play; infrastructure rather than product, but credible.
Team capacity drops to 1–2 engineers	Scope Aegis to the vault + solvency attestation	Smaller (no Aegis, no PMW) on FCC + FDC + FAssets alone.
No institutional LOI achievable	Reframe Aegis for the retail XRP whale segment	Looses some institutional flare but retains the technical story.
A competing team submits a near-ideal concept	Pivotal to Concept 4: FXRP stablecoin issuer	Differs from Aegis in that it is a stablecoin primitive while retaining FCC + AI.

16. Final Recommendation

Build **Aegis**.

Aegis is a verifiable, confidential, AI-managed cross-chain treasury protocol for XRP-native institutions on Flare. It uses every enshrined Flare primitive in a load-bearing way — FAssets for asset onboarding, FTSO for price discovery, FDC for external state attestation, FCC for confidential position computation and verifiable AI inference, and PMW for cross-chain execution. It targets a real and growing customer segment — corporate XRP treasuries, validated by VivoPower's USD 100 million commitment — that no current Flare product serves. It has a clear business model, a credible post-hackathon trajectory, and a demo that dramatises the confidentiality-to-verifiability transformation at the heart of the Flare 2.0 thesis.

The recommendation rests on three converging facts that are unlikely to change before the submission deadline. First, the judging rubric weights Flare integration quality above all else, and Aegis is the only concept in the top fifty that uses all five Flare primitives load-bearing. Second, Flare's strategic priorities — institutional onboarding, verifiable AI, and cross-chain execution — map exactly onto Aegis's three core capabilities. Third, the competitive landscape is fragmented: most entries will target retail onboarding or simple FAssets wrappers, leaving the institutional + multi-primitive space open.

The principal risks are PMW integration (mitigated by week-1 validation and a simulated fallback), institutional pilot acquisition (mitigated by outreach and a fallback narrative), and demo reliability (mitigated by four rehearsals and a pre-recorded backup). None of these risks is unmanageable, and the upside — a Grand Prize, a reference implementation of Flare 2.0, and a fundable startup — is substantial.

Execute the day-by-day plan in Section 10. Validate PMW on Coston2 in week one. Cut features ruthlessly if behind, in the priority order: SDK first, then multi-vault, then auditor dashboard, then the ML component. Rehearse the demo four times. Secure at least one institutional letter of interest. Submit at least 24 hours before the deadline.

This is the project with the highest probability of winning the Grand Prize of the Flare Summer Signal hackathon.

Aegis — A Verifiable, Confidential, AI-Managed Cross-Chain Treasury and Autonomous Risk Layer for XRP-Native Institutions on Flare.

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